NATIONAL JUNIOR COLLEGE

SENIOR HIGH 2 PRELIMINARY EXAM

Higher 2

CANDIDATE NAME

SUBJECT CLASS

REGISTRATION NUMBER

PHYSICS

9749/01

Paper 1 Multiple Choice

24 September 2020

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTION FIRST

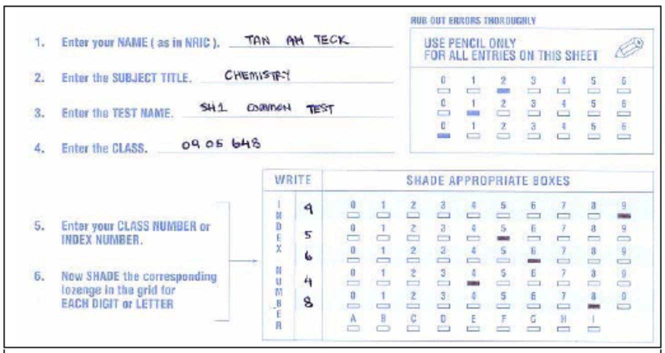
Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your name, subject class and registration number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **THIRTY** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

INSTRUCTIONS ON SHADING OF REGISTRATION NUMBER



Shade the index number in a 5 digit format (12345) on the Answer Sheet.

OAS index number is in 5-digit format.

5 digit format: **2nd digit** and the **last four digits** of the Reg Number.

Note: Q10 is void

This document consists of **16** printed pages and **0** blank page.

[Turn over

Data

speed of light in free space	$c = 3.00 \times 10^{8}\,\mathrm{m\,s^{-1}}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
	$(1/(36\pi)) \times 10^{-9}\,\mathrm{F\,m^{-1}}$
elementary charge	$e = 1.60 \times 10^{-19}\,\mathrm{C}$
the Planck constant	$h = 6.63 \times 10^{-34}\,\mathrm{J\,s}$
unified atomic mass constant	$u = 1.66 \times 10^{-27}\,\mathrm{kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31}\,\mathrm{kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27}\,\mathrm{kg}$
molar gas constant	$R = 8.31\,\mathrm{J\,K^{-1}\,mol^{-1}}$
the Avogadro constant	$N_A = 6.02 \times 10^{23}\,\mathrm{mol^{-1}}$
the Boltzmann constant	$k = 1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
acceleration of free fall	$g = 9.81\,\mathrm{m\,s^{-2}}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
work done on / by a gas	$W = p\Delta V$
hydrostatic pressure	$p = \rho gh$
gravitational potential	$\phi = -Gm/r$
temperature	$T/\mathrm{K} = T/°\mathrm{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V} < c^2 >$
mean translational kinetic energy of an ideal gas molecule	$E = \frac{3}{2}kT$
displacement of particle in s.h.m.	$x = x_0 \sin\omega t$
velocity of particle in s.h.m.	$v = v_0 \cos\omega t$ $= \pm\,\omega\sqrt{x_0^2 - x^2}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$1/R = 1/R_1 + 1/R_2 + \ldots$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	$x = x_0 \sin\omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
radioactive decay	$x = x_0 \exp(-\lambda t)$
decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

1. The speed of a wave in deep water depends on its wavelength L and the acceleration of free fall g.

What is a possible equation for the speed v of the wave?

A $v = \frac{2\pi g}{L}$ **B** $v = \frac{gL}{4\pi^2}$ **C** $v = 2\pi\sqrt{\frac{g}{L}}$ **D** $v = \sqrt{\frac{gL}{2\pi}}$

2. The power P dissipated in a resistor of resistance R is calculated using the expression

$$P = \frac{V^2}{R}$$

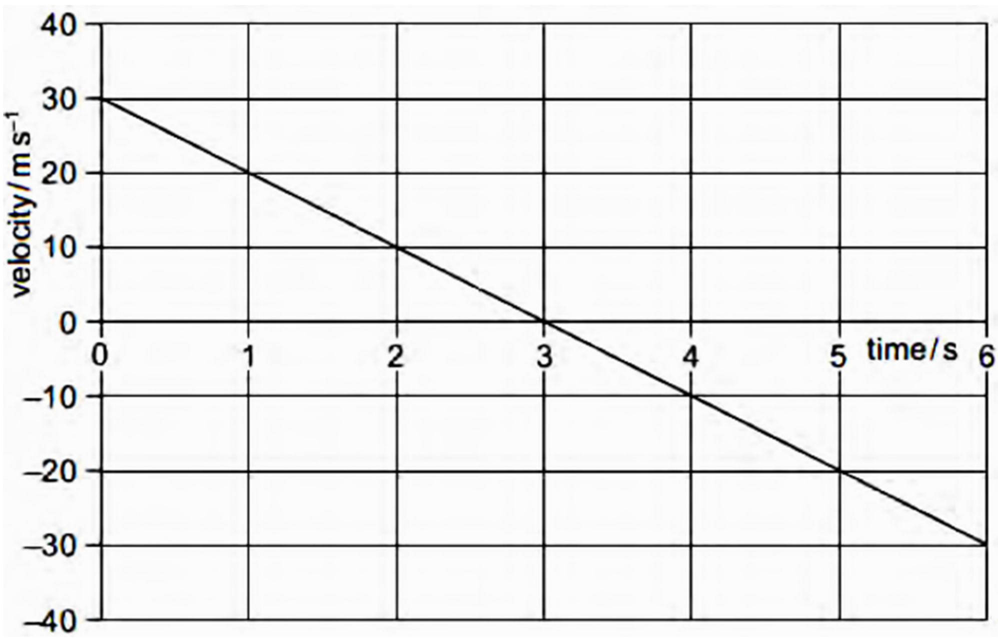
where V is the potential difference across the resistor.

In an experiment to determine P, R is measured to within a percentage uncertainty of 2%.

What is the maximum percentage uncertainty in the measurement of V such that P can be determined with a maximum uncertainty of $\pm$ 5%?

A 1.5% **B** 3.0% **C** 3.5% **D** 7%

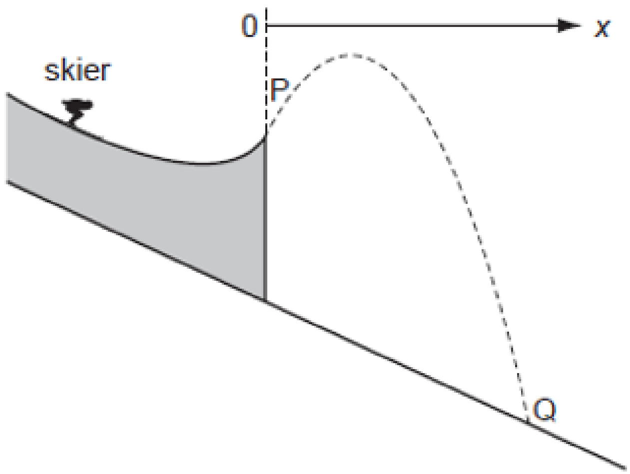
3. A stone is thrown vertically upwards. A student plots the variation with time of its velocity.



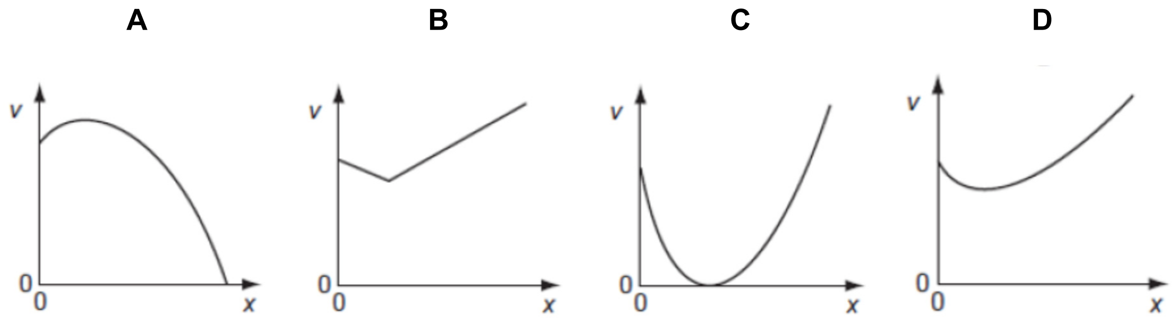
What is the vertical displacement of the stone from its starting point after 5 seconds?

A 20 m **B** 25 m **C** 45 m **D** 65 m

4. The dotted line shows the path of a competitor in a ski-jumping competition.



Ignoring air resistance, which graph best represents the variation of his speed v with the horizontal distance x covered from the start of his jump at P before landing at Q?



5. Two magnets P and Q, having masses M_P and M_Q respectively, exert forces on each other and have no other external forces acting on them. The force acting on P is F, which gives P an acceleration a.

Which of the following pairs is correct?

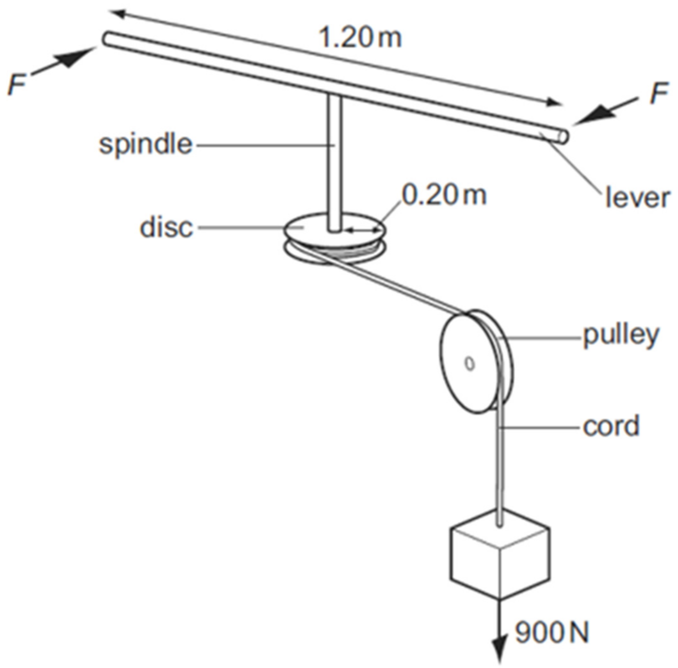
	Magnitude of force on Q	Magnitude of acceleration of Q
A	$\frac{M_Q}{M_P}F$	a
B	$\frac{M_P}{M_Q}F$	a
C	F	$\frac{M_Q}{M_P}a$
D	F	$\frac{M_P}{M_Q}a$

6. Particles X (of mass 4 units) and Y (of mass 9 units) move directly towards each other, collide and then separate.

If Δv_x is the change in velocity of X and Δv_y is the change in velocity of Y, the magnitude of ratio $\frac{\Delta v_x}{\Delta v_y}$ is

A 9 / 4 **B** 3 / 2 **C** 1 **D** 2 / 3

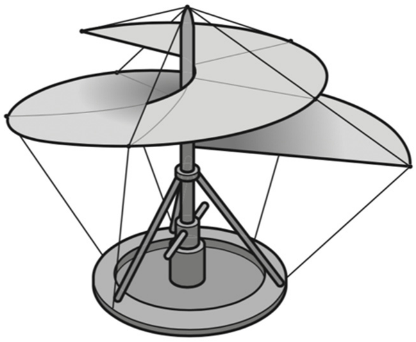
7. A spindle is attached at one end to the centre of a lever 1.20 m long and at its other end to the centre of a disc of radius 0.20 m. A cord is wrapped round the disc, passes over a pulley and is attached to a 900 N weight.



What is the minimum force F, applied to each end of the lever, that could lift the weight?

A 75 N **B** 150 N **C** 300 N **D** 950 N

8. Leonardo da Vinci proposed a flying machine that would work like a screw to lift the pilot into the air. The 'screw' is rotated by the pilot.

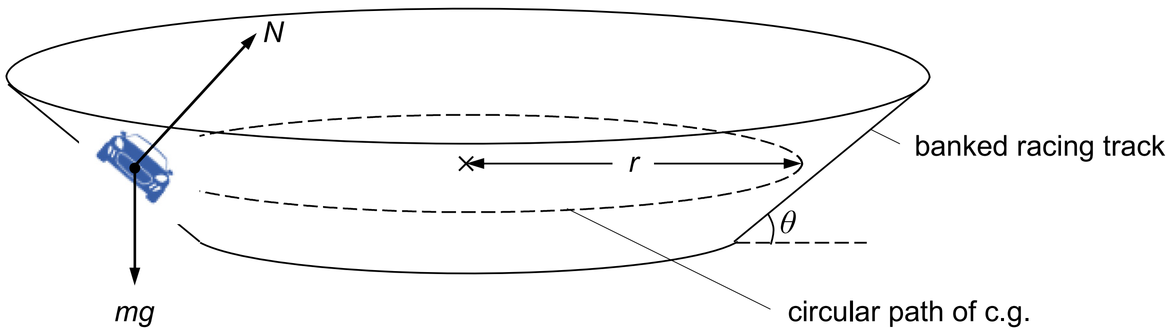


The machine and the pilot together have a total mass of 120 kg.

Which useful output power must the pilot provide to move vertically upwards at a constant speed of 2.5 m s^{-1}?

A 48 W **B** 300 W **C** 470 W **D** 2900 W

9. A racing car of mass *m* is moving with speed *v* on a banked racing track. The angle of inclination of the track is *θ*. The centre of gravity (c.g.) of the car moves along a circular path of horizontal radius *r*.



The centripetal force arises entirely from a component of the normal reaction *N* from the track.

Which of the following relations is correct?

A $N \sin\theta = mg$

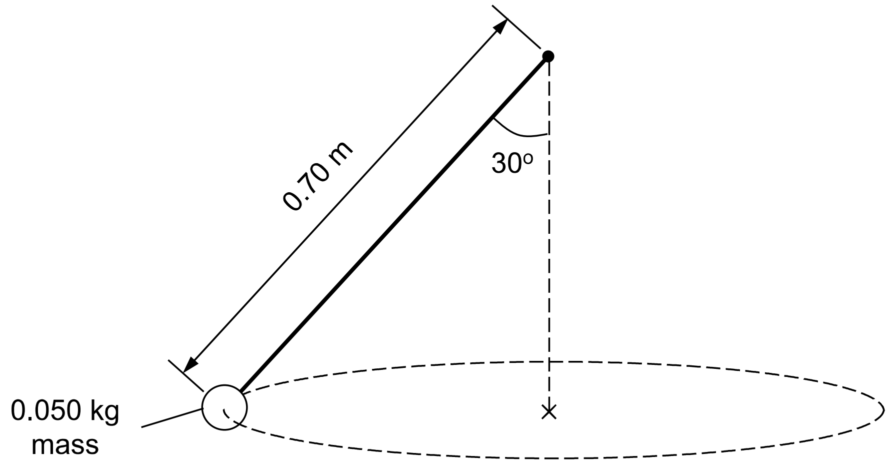
B $N = mg \cos\theta$

C $rg \tan\theta = v^2$

D $v^2 \tan\theta = rg$

~~10.~~ ~~A mass of 0.050 kg is attached to one end of a piece of elastic cord of unstretched length 0.50 m. The elastic cord obeys Hooke's Law and its force constant is 40 N m⁻¹.~~

~~The mass is rotated in a horizontal circular path as shown in the diagram.~~



~~The length of the cord is 0.70 m.~~

~~What is the speed of the mass?~~

A 5.3 m s^{-1} **B** 7.5 m s^{-1} **C** 9.2 m s^{-1} **D** 28 m s^{-1}

11. A planet of mass *M* and radius *R* rotates so rapidly that loose material at the equator just remains on the surface. What is the period of rotation of the planet?

A $2\pi\sqrt{\frac{R}{GM}}$ **B** $2\pi\sqrt{\frac{R^2}{GM}}$ **C** $2\pi\sqrt{\frac{GM}{R^3}}$ **D** $2\pi\sqrt{\frac{R^3}{GM}}$

12. The following data refer to two planets.

	radius / km	density / kg m^{-3}
Planet P	2000	6000
Planet Q	16000	3000

The gravitational field strength at the surface of P is 13.4 N kg^{-1}.

What is the gravitational field strength at the surface of Q?

A 3.4 N kg^{-1} **B** 13.4 N kg^{-1} **C** 53.6 N kg^{-1} **D** 80.4 N kg^{-1}

13. In a mixture of two monatomic ideal gases X and Y in thermal equilibrium, a molecule of Y has twice the mass of a molecule of X. The mean random translational kinetic energy of the molecules of Y is 6.0×10^{-21} J.

What is the mean random translational kinetic energy of the molecules of X?

A 3.0×10^{-21} J

B 4.2×10^{-21} J

C 6.0×10^{-21} J

D 12×10^{-21} J

14. The specific latent heat of vaporization of water at 20°C is appreciably greater than the value of 100°C.

This is because

A The molecules in the liquid are more tightly bound to one another at 20°C than at 100°C.

B More work must be done in expanding the water vapour against atmospheric pressure at 20°C than at 100°C.

C The root mean square speed of the vapour molecules is less at 20°C than at 100°C.

D The specific latent heat at 20°C includes the energy necessary to raise the temperature of one kilogram of water from 20°C to 100°C.

15. An ideal gas, contained in a cylinder by a frictionless piston, is allowed to expand from volume V_1, at a pressure p_1, to volume V_2, at pressure p_2. Its temperature is kept constant throughout. The work done by the gas is

A Zero, because it obeys Boyle's Law and therefore $p_2V_2 - p_1V_1 = 0$.

B Negative, because the pressure has decreased and so the force on the piston has been diminishing.

C Zero, because it has been kept at constant temperature and so its internal energy is unchanged.

D Positive, because the volume has increased.

16. A body moves with simple harmonic motion of amplitude A and frequency $\frac{b}{2\pi}$.

What is the magnitude of the acceleration when the body is at maximum displacement?

A zero **B** $4\pi^2Ab^2$ **C** Ab^2 **D** $\frac{4\pi^2A}{b^2}$

17. Which one of the following statements concerning forced vibrations and resonance is correct?

A An oscillating body that is not resonating will return to its natural frequency when the forcing vibration is removed.

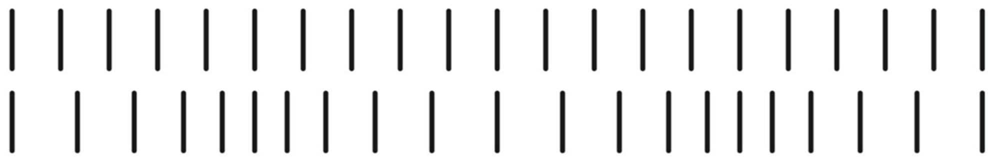
B At resonance, the displacement of the oscillating body is 180° out of phase with the forcing vibration.

C A pendulum with a dense bob is more heavily damped than one with a less dense bob of the same size.

D Resonance can only occur in mechanical systems.

18. The top row of bars represents a set of particles inside the Earth and at rest.

The lower row represents the same particles at one instant as a longitudinal wave passes from left to right through the Earth.



What should be measured to determine the amplitude of the oscillations of the particles in the lower row as the wave passes?

A half the maximum displacement of the particles from their position at rest

B half the maximum distance apart of the particles

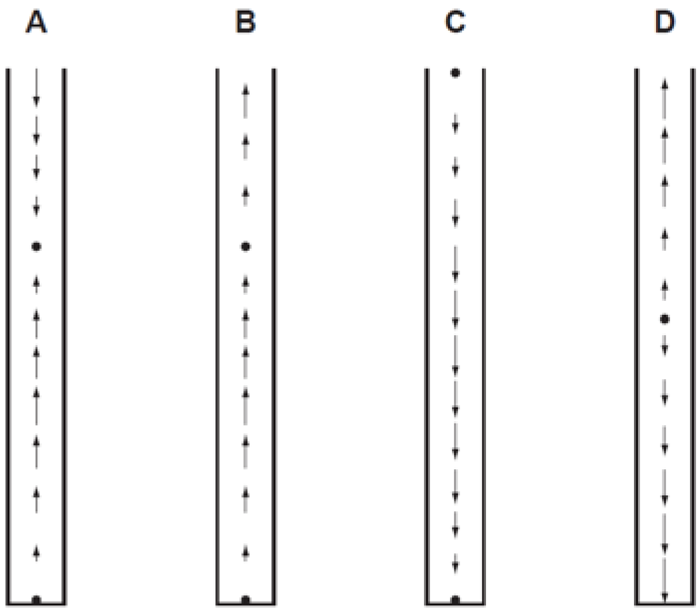
C the maximum displacement of the particles from their position at rest

D the maximum distance apart of the particles

19. A stationary longitudinal wave is set up in a pipe.

In the diagram below, the length of each arrow represents the amplitude of the motion of the air molecules, and the arrow head shows the direction of motion at a particular instant.

Which diagram shows a stationary wave in which there are two nodes and two antinodes?

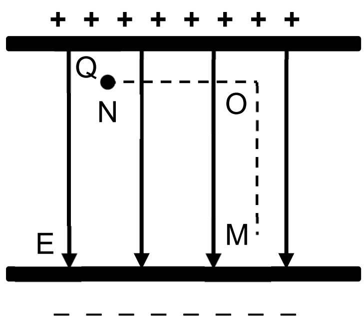


20. Light of wavelength 720 nm from a laser X is incident normally on a diffraction grating and a diffraction pattern is observed. Light from a laser Y is then also incident normally on the same grating. The third-order maximum due to laser Y is seen at the same place as the second-order maximum due to laser X.

What is the wavelength of the light from laser Y?

A 480 nm **B** 540 nm **C** 720 nm **D** 1080 nm

21. The figure below shows a charged particle Q in a uniform electric field E.



Which of the following statements about Q is correct?

A No work is done when the positive charge moves from N to O then to M.

B No work is done when the positive charge moves from O to M only.

C Work is done by field when the positive charge moves from M to O, but work is done by external force when it moves from O to M.

D Work done on the positive charge when it moves from N to O to M is the same as when it moves from N to M directly.

22. Which one of the following statements about *electric field strength* and *electric potential* is **incorrect**?

A Electric potential is a scalar quantity.

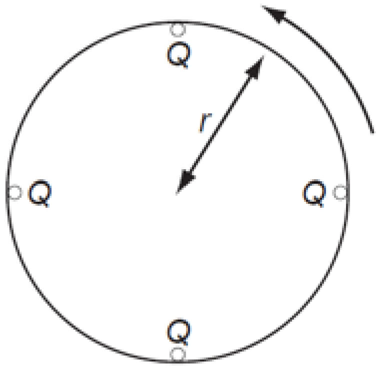
B Electric field strength is a vector quantity.

C Electric potential is zero whenever the electric field strength is zero.

D The potential gradient is proportional to the electric field strength.

23. Four point charges, each of charge Q, are placed on the edge of an insulating disc of radius r.

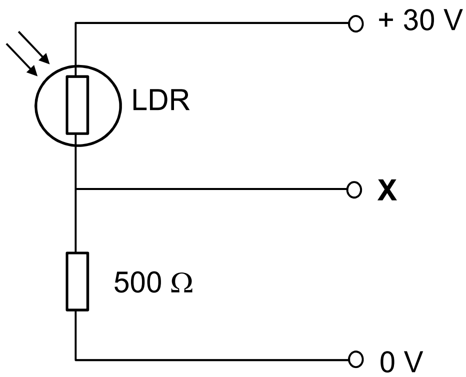
The frequency of rotation of the disc is f.



What is the equivalent electric current at the edge of the disc?

A $4Qf$ **B** $\frac{4Q}{f}$ **C** $8\pi rQf$ **D** $\frac{2Qf}{\pi r}$

24. The light dependent resistor (LDR) and a 500 Ω resistor form a potential divider between voltage lines held at +30 V and 0 V as shown.



The resistance of the LDR is 1000 Ω in the dark but drops to 100 Ω in bright light.

What is the corresponding change in the potential at **X**?

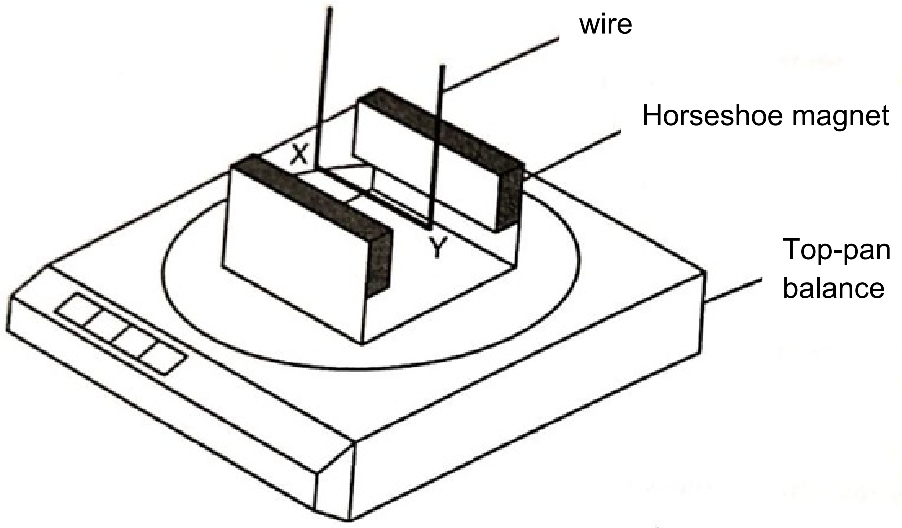
A A rise of 25 V

B A rise of 15 V

C A fall of 15 V

D A fall of 25 V

25. A horseshoe magnet rests on a top-pan balance with a wire situated between the poles of the magnet.



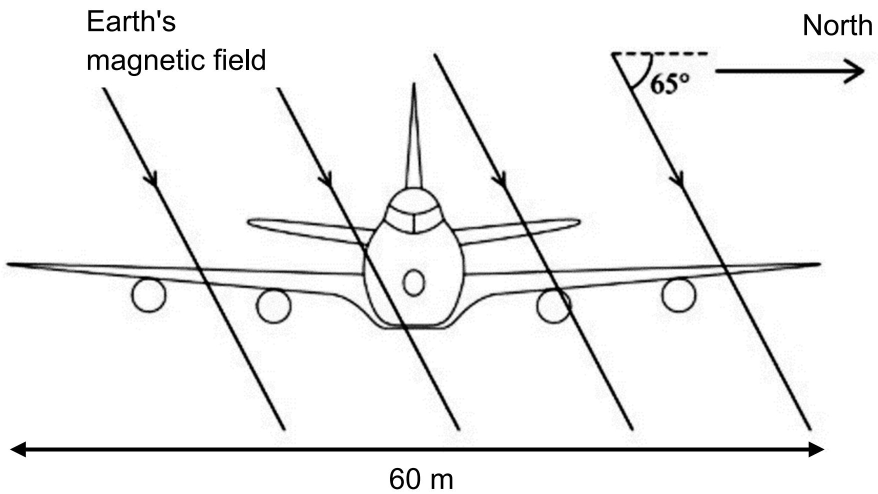
With no current in the wire, the reading on the balance is 142.0 g.

With a current of 2.0 A in the wire in the direction XY, the reading on the balance changes to 144.6 g.

What is the reading on the balance when there is a current of 3.0 A in the wire in the direction YX?

A 138.1 g **B** 140.7 g **C** 145.9 g **D** 148.5 g

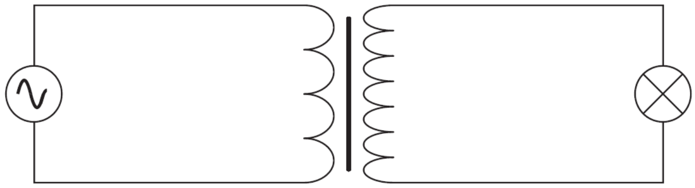
26. An aircraft, of wing span 60 m, flies horizontally at a speed of 150 m s^{-1}. The Earth's magnetic field in the region of the plane is 1.0×10^{-5} T and at an angle of 65° to the horizontal as shown below.



What emf is induced across the wing tips of the plane?

A 0.038 V **B** 0.082 V **C** 0.090 V **D** 9.0 V

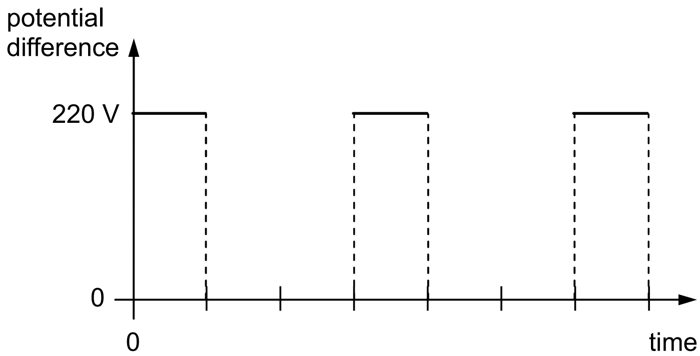
27. The primary coil of a step-up ideal transformer is connected to a source of alternating potential difference. The secondary coil is connected to a lamp.



Which line, **A** to **D**, in the table correctly describes the flux linkage and current through the secondary coil in relation to the primary coil?

	$\frac{\text{secondary magnetic flux linkage}}{\text{primary magnetic flux linkage}}$	$\frac{\text{secondary current}}{\text{primary current}}$
A	> 1	< 1
B	< 1	< 1
C	> 1	> 1
D	< 1	> 1

28. An alternating current is supplied to a rectifier. The output from the rectifier has the shape shown in the diagram, with a maximum value of 220 V.



This output is used as the supply to a resistor of resistance 60.0 Ω.

What are the r.m.s. values of the current and the power to the resistor?

	r.m.s. current / A	power / W
A	1.22	89.6
B	1.83	202
C	2.12	269
D	2.59	403

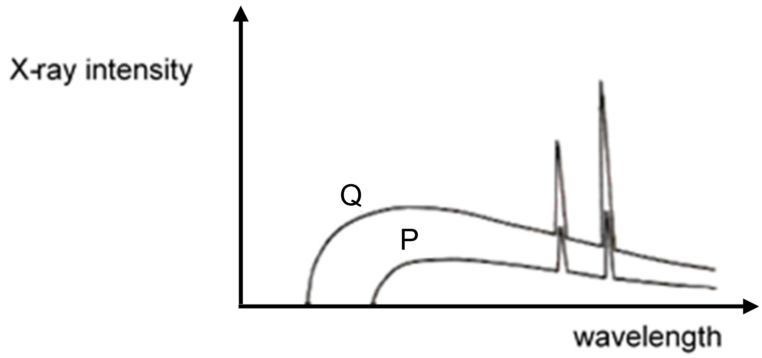
29. When a beam of light of wavelength λ is incident normally on a plane mirror, it is reflected totally.

The intensity of the light corresponds to a rate of n photons per second.

What is the force exerted on the mirror by the beam of light?

A $nh\lambda$ **B** $\dfrac{nh}{\lambda}$ **C** $2nh\lambda$ **D** $\dfrac{2nh}{\lambda}$

30. The given graph show the variation of X-ray intensity with wavelength emitted from two X-ray tubes P and Q.



Which of the following could be a deduction from the graphs?

A X-ray tube Q has the higher voltage applied to it and the target material in both tubes is the same.

B X-ray tube Q has the higher voltage applied to it and the target material in both tubes are different.

C X-ray tube P has the higher voltage applied to it and the target material in both tubes is the same.

D X-ray tube P has the higher voltage applied to it and the target material in both tubes are different.

NATIONAL JUNIOR COLLEGE

SENIOR HIGH 2 PRELIMINARY EXAMINATION

Higher 2

CANDIDATE NAME

SUBJECT CLASS

REGISTRATION NUMBER

PHYSICS **9749/02**

Paper 2 Structured Questions

Candidate answers on the Question Paper.

1 September 2020

2 hours

No Additional Materials are required.

READ THE INSTRUCTION FIRST

Write your subject class, registration number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

The use of an approved scientific calculator is expected, where appropriate.

Answers **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	/ 10
2	/ 10
3	/ 9
4	/ 10
5	/ 10
6	/ 9
7	/ 22
Total (80m)	

This document contains **23** printed pages and **1** blank page.

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molar gas constant	$R = 8.31\,\mathrm{J\,K^{-1}\,mol^{-1}}$
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hydrostatic pressure	$p = \rho gh$
gravitational potential	$\phi = -Gm/r$
temperature	$T/\text{K} = T/{}^\circ\text{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V} < c^2 >$
mean translational kinetic energy of an ideal gas molecule	$E = \frac{3}{2}kT$
displacement of particle in s.h.m.	$x = x_0 \sin\omega t$
velocity of particle in s.h.m.	$v = v_0 \cos\omega t$ $= \pm\,\omega\sqrt{{x_0}^2 - x^2}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$1/R = 1/R_1 + 1/R_2 + \ldots$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	$x = x_0 \sin\omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
radioactive decay	$x = x_0 \exp(-\lambda t)$
decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

[Turn over

1 **(a)** State the *principle of moments*.

....

.... [1]

(b) It is said that Archimedes used huge levers to sink Roman ships invading the city of Syracuse. A possible system is shown in Fig. 1.1 where a rope is hooked on to the front of the ship and the lever is pulled by several men.

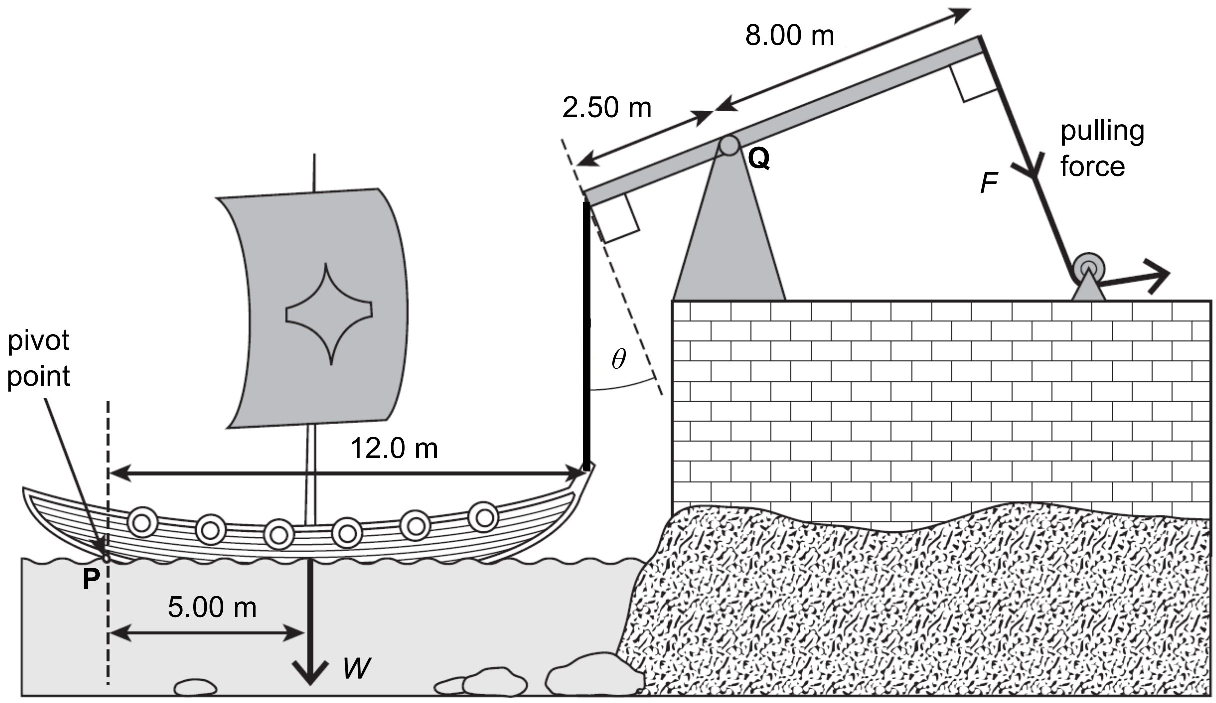


Fig. 1.1

When the ship is just lifted from the water, upthrust is negligible.
The weight W of a Roman ship was 3.40×10^4 N.
The rope hooked to the front of the ship is vertical. The angle θ is 24.0°

The centre of gravity of the light beam of the lever is at point **Q**.

(i) Assume the ship pivots about point **P**. Show that

1. the tension in the rope hooked to the front of the ship is 14.2 kN,

[1]

2. the force F is 4.05 kN.

[2]

(ii) A force R is exerted on the beam of the lever at point **Q** at an angle α to the vertical, as shown in Fig. 1.2.

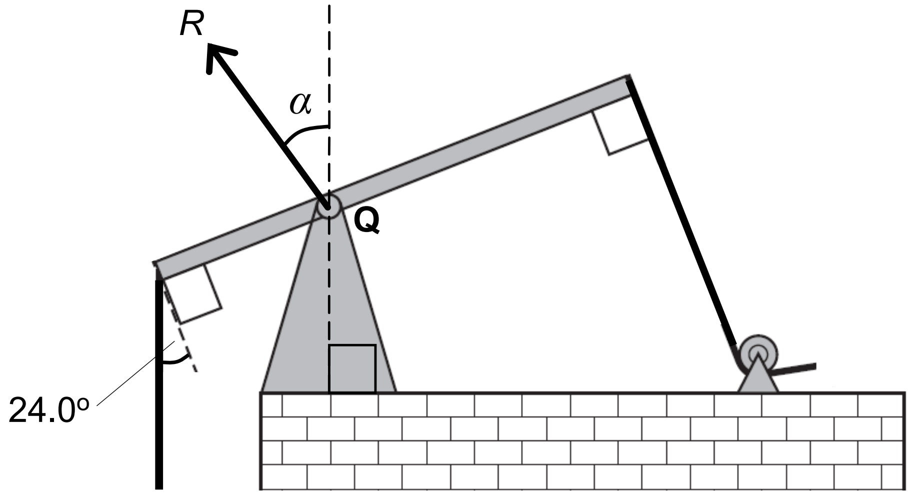


Fig. 1.2

Use your answer in **(i)** to determine the magnitude of R and the angle α.

magnitude of force = ... N

angle α = ... °

[3]

(c) Fig. 1.2 shows a trailer attached to the towbar of a stationary car.

All forces acting on the stationary trailer are vertical. The weight W of the trailer acts through its centre of gravity. F is the force exerted by the towbar on the trailer. F_R is the **total** normal reaction force experienced by the trailer.

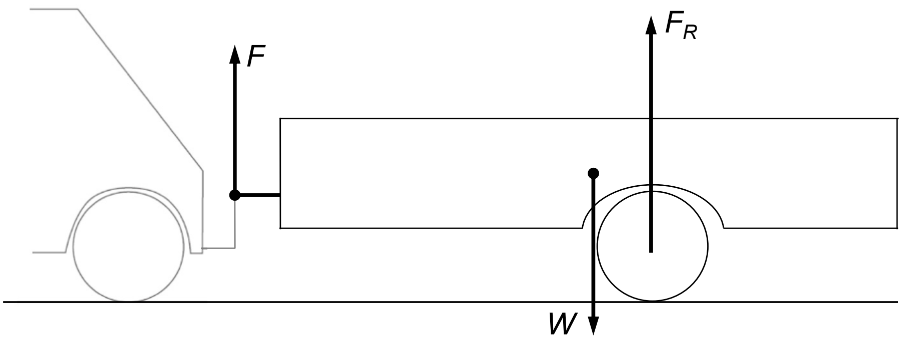


Fig. 1.2

State and explain the changes to the magnitude and direction of the force F when the car starts to move forwards.

....

....

....

....

....

.... [3]

[Total: 10]

Question 2 begins over the page.

[Turn over

2 The space race started in 1957, the year that the Soviet Union put the first artificial satellite, Sputnik 1 in orbit around the Earth. Sputnik 1 was launched using a rocket similar to the one shown in Fig. 2.1.

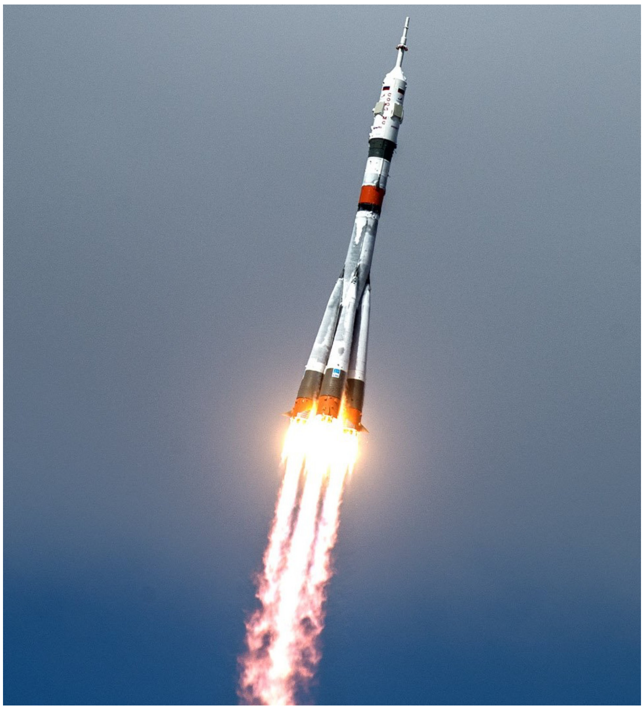


Fig. 2.1

The rocket was propelled by hot gases from the burning of a mixture of liquid oxygen and kerosene.

(a) Using momentum consideration, explain how the ejection of hot gases can result in a forward thrust on the rocket.

....

....

....

.... [2]

(b) Explain why the resultant force on the rocket in a vacuum is larger than when it is in air.

....

....

....

.... [2]

(c) Fig. 2.2 shows a rocket in a vertical take–off position.

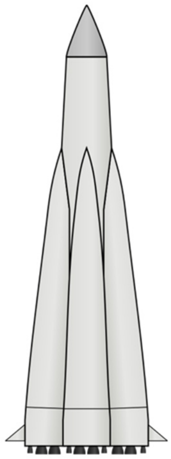


Fig. 2.2

The rocket, with its fuel, has a total mass of 1.2×10^6 kg.
Immediately after a vertical take–off, its acceleration is 3.5 m s^{-2}.

Assume that air resistance is negligible.

(i) On Fig. 2.2, draw arrows to show the forces acting on the rocket immediately after take-off. Label the forces. [2]

(ii) Determine the thrust of the rocket motors immediately after take-off.

thrust = ………………………………… N [2]

(iii) 1.0 minute after the take–off, the rocket is still rising vertically. The thrust remains constant and its acceleration is 18.7 m s^{-2}.

Determine the average rate of fuel consumption during this 1.0 minute interval.

rate of fuel consumption = ………………………………… kg s^{-1} [2]

[Total: 10]

3 **(a)** On one particular ride in an amusement park, passengers in a carriage car 'loop–the–loop' in a vertical circle, as illustrated in Fig. 3.1.

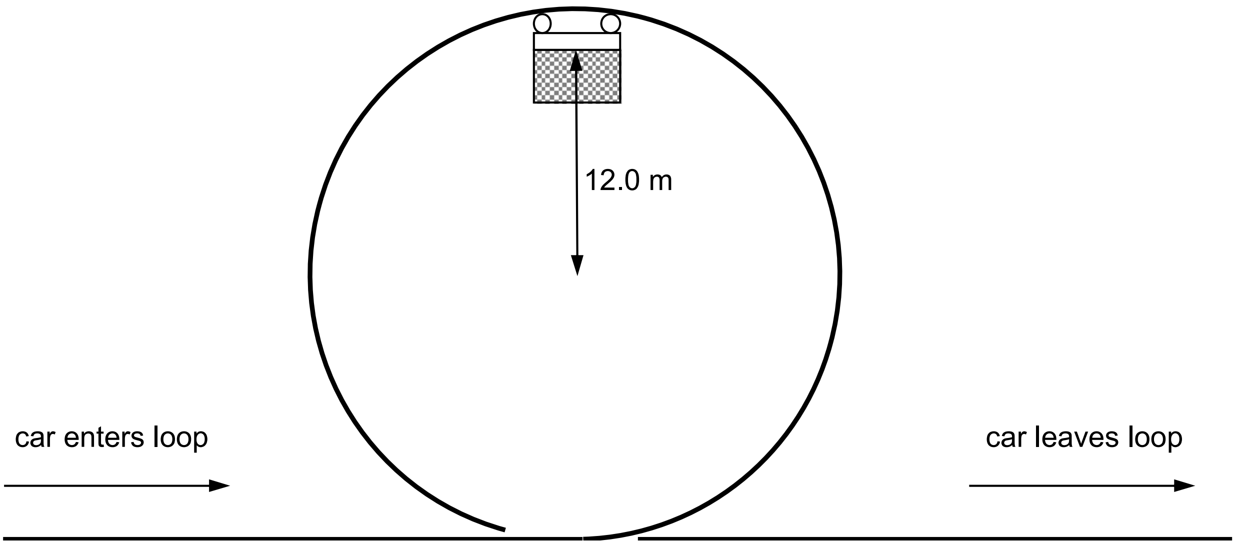


Fig. 3.1

The car moves freely, so there is no frictional forces and no driving force propelling the car when it is moving in the loop.

The centre of gravity of the car moves in a circle with radius 12.0 m.

Determine

(i) the minimum speed of the carriage car at the top of the circular track shown in Fig. 3.1,

minimum speed = ... m s^{-1} [3]

(ii) the minimum speed of the car leaving the loop.

minimum speed = ... m s^{-1} [2]

(b) In another ride in the same amusement park, passengers sit in carriages which travel at variable angular velocities in a vertical circle. Fig. 3.2 shows the motion of one of the carriages.

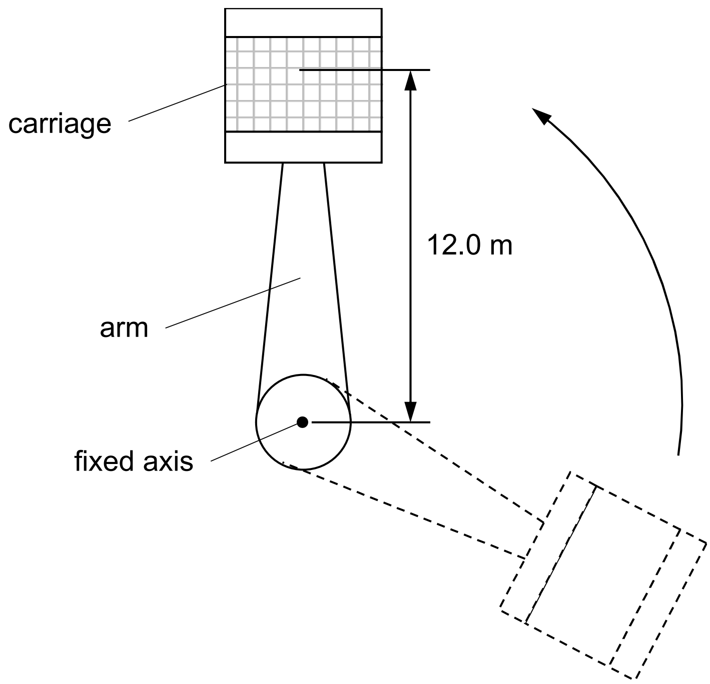


Fig. 3.2

The centre of gravity of the carriage moves in a circle with radius 12.0 m.

(i) State what is meant by the *angular velocity*.

....

....

....[2]

(ii) Suggest why it is possible for the carriage in Fig. 3.2 to be at rest (i.e. zero angular velocity) at the top of the circle whereas it is impossible for the car in **(a)**.

....

....

....

....[2]

[Total: 9]

4 **(a)** Define *potential* at a point in an electric field.

....

.... [1]

(b) Fig. 4.1. shows the equipotential lines of a non-uniform electric field.

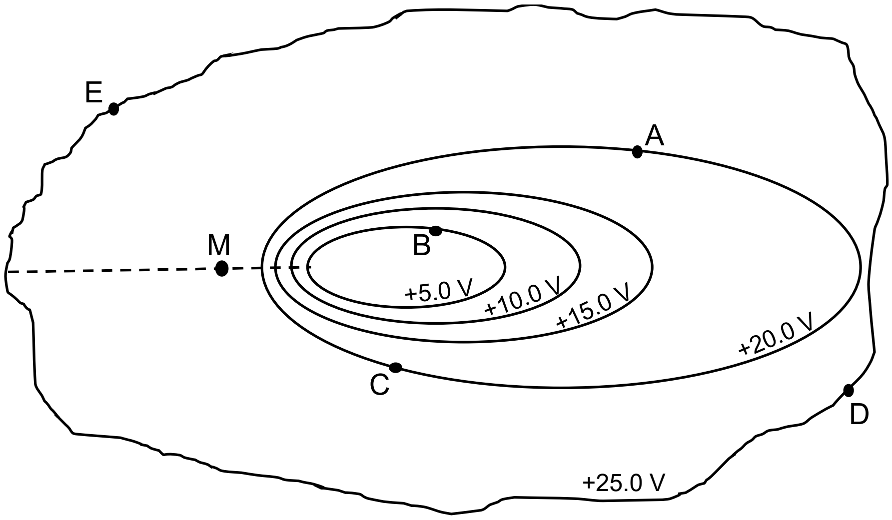


Fig 4.1

(i) State the potential difference between point A and point B.

potential difference = ... V [1]

(ii) Determine the work done to bring a charge of +5.0 μC from point B to point E, via point C and point D.

work done = ... J [2]

(iii) Explain whether the work done you have obtained in **(ii)** is the work done by the electric field or the work done by an external agent.

....

....

....

....[2]

(c) **(i)** State the relation between electric field strength and potential.

……………………………………………………………………………………………………..

…………………………………………………………………………………………………….[1]

(ii) Draw an arrow at point **D** in Fig. 4.1 to show the direction of the electric field strength at that point. [1]

(ii) Another positive charge in released from point M in Fig. 4.1. It moves along the dotted line.

Describe, without further calculation, the motion of this positive charge along the dotted line.

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

…………………………………………………………………………………………………….[2]

[Total: 10]

5 **(a)** Define *magnetic flux density*.

....

....

.... [1]

(b) Electrons enter a rectangular block of metal at right-angles to face PQFE, as shown in Fig. 5.1.

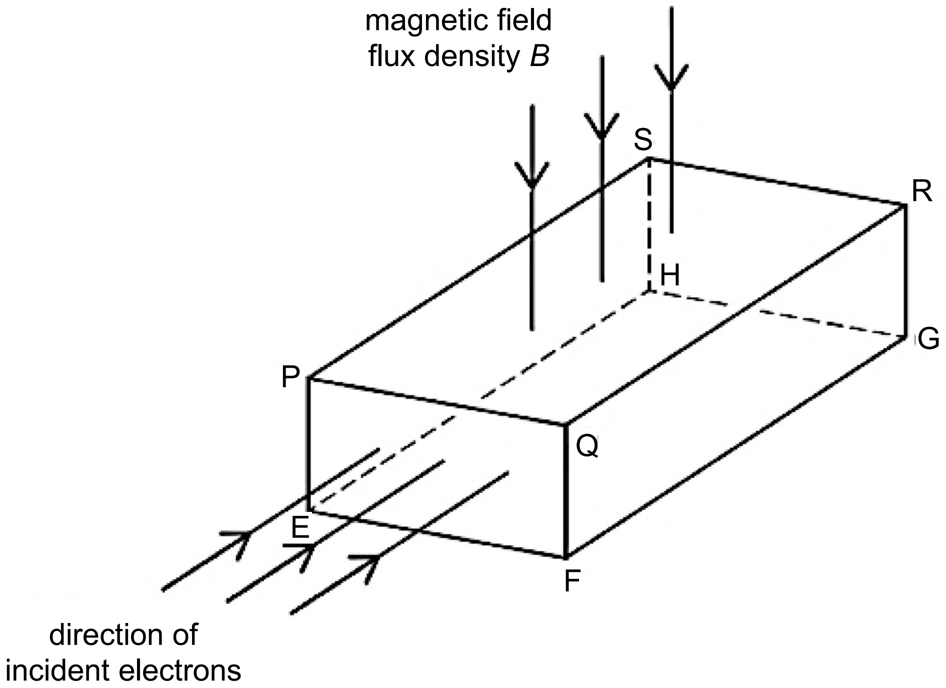


Fig. 5.1

A uniform magnetic field of flux density B is directed at right-angles to face PQRS.

The electrons each have a charge $-q$ and drift speed v in the metal.

Explain why an electron experiences a force due to the magnetic field. State the direction of this force when the electron first enters the block of metal.

....

....

....

.... [2]

(c) After some time, a potential difference is established across two opposite faces of the metal in **(b)** causing the electrons to travel in a straight line through the metal.

(i) Explain why a potential difference is established across two opposite faces.

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………. [2]

(ii) State the two faces between which the potential difference is established.

face …………………………and face…………………………. [1]

(d) The potential difference V_H established across the metal in **(c)** is given by the expression

$$V_H = \frac{BI}{ntq}$$

where I is the current in the metal, B is the magnetic flux density, n is the number of electrons per unit volume, q is the electronic charge and t is the thickness of the metal

(i) Aluminium has a density of 2.7 g cm^{-3}.
The mass of one atom of Aluminium is 4.5×10^{-26} kg.
Assume there is one free electron available to carry charge per atom of aluminium.

Show that the number of charge carriers per unit volume in aluminium is 6.0×10^{28} m^{-3}.

[2]

(ii) An aluminium foil has a thickness t of 0.090 mm. The current in the foil is 4.6 A.

A uniform magnetic field of flux density 0.15 T acts at right-angles to the foil.

Calculate the potential difference V_H that is generated.

V_H = ………………………………… V [2]

[Total: 10]

6 A set of 12 coloured filament light bulbs is designed for use with a 24 V d.c. supply. The 12 light bulbs are connected in series, as shown in Fig. 6.1.

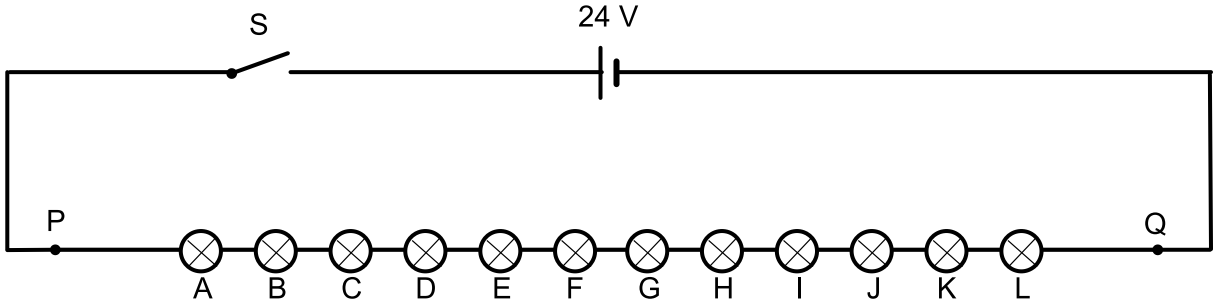


Fig. 6.1

The d.c. voltage supply has negligible internal resistance and each of the 12 light bulbs has the same constant electrical resistance.

(a) When switch S is closed, the total power dissipated across the 12 light bulbs is 6.0 W.

Determine

(i) the current passing through each light bulb,

current = ... A [1]

(ii) the potential difference across each light bulb,

potential difference = ... V [1]

(iii) the resistance of each light bulb

resistance = ... Ω [1]

(b) In another situation, the light bulbs do not light up when switch S is closed. The fault is due to a single break in connection (open circuit) somewhere in the circuit.

A voltmeter is used to locate where this break in connection occurs.

Each of the following observations are **independent** of the other. State the possible location of the break in connection for each observation.

(i) The potential difference between P and Q is zero.

……………………………………………………………………………………………………..

…………………………………………………………………………………………………… [1]

(ii) The potential difference is zero across every light bulb except H. The potential difference across H is 24 V.

……………………………………………………………………………………………………..

…………………………………………………………………………………………………… [1]

(iii) The potential difference between P and Q is 24 V but the potential difference is zero across every single light bulb.

……………………………………………………………………………………………………..

…………………………………………………………………………………………………… [1]

(c) A certain type of light bulb is designed such that the resistance of the bulb drops to zero when the filament fails. The light bulbs in Fig. 6.1 are replaced by this type of light bulb.

The filament of one light bulb fails when switch S is closed.

Determine

(i) the potential difference across each of the remaining light bulb,

potential difference = ………………………………… V [1]

(ii) the fractional increase in the power dissipated in each bulb.

fractional increase = ………………………………… [2]

[Total: 9]

7 Read the following article and answer the questions that follow.

Anti-reflective coating

When light is incident normally at the interface between a medium with refractive index n_1 and a second medium with refractive index n_2, both reflection and transmission of the light may occur, as shown in Fig. 7.1.

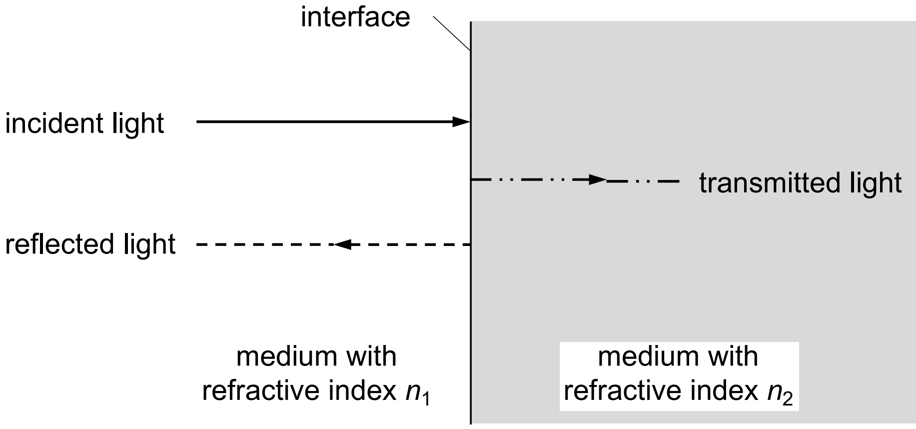


Fig. 7.1

The ratio r of the reflected wave amplitude to the incident wave amplitude is given by

$$r = \frac{n_2 - n_1}{n_2 + n_1}$$

However, these amplitudes can be complex numbers. Therefore, it is more convenient to use the reflectance R and transmittance T where

$$R = \frac{\text{reflected power}}{\text{incident power}}$$

and $R + T = 1$.

Anti-reflective (AR) coating was discovered in 1886 by British Physicist Lord Rayleigh. Upon testing some old, tarnished glass pieces (chemical reactions between the environment and the optical glass of his time tended to cause surface tarnishing on glass as it aged), Lord Rayleigh found that – to his surprise – these tarnished pieces transmitted more light than clean, new pieces.

Today, AR coating is used on eyeglasses to improve vision and reduces eye strain due to the ability of AR coating to reduce reflections from the front and back surfaces of the eyeglass lenses and allow more light to pass through the lenses. Fig. 7.2 shows the reflection of light from the lenses without and with AR coating.

(a) eyeglasses without AR coating

(b) eyeglasses with AR coating

Fig. 7.2

AR coating is especially beneficial when used on high refractive index lenses, which reflect more light than regular plastic lenses. Generally, the higher the refraction index of the lens material, the more light that will be reflected from the surface of the lens. This can be particularly troublesome in low-light conditions, such as when driving at night.

The reflection of a monochromatic light incident normally on a AR coated lens is minimised when the thickness of the coating is about a quarter of the wavelength, as shown in Fig. 7.3.

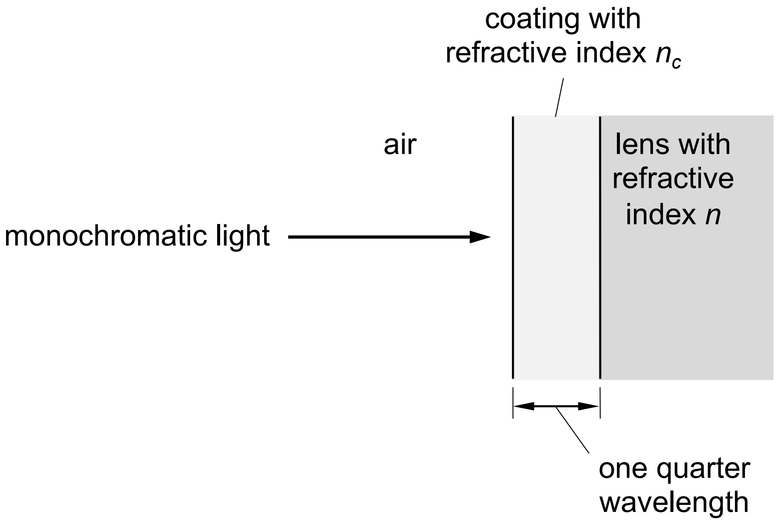
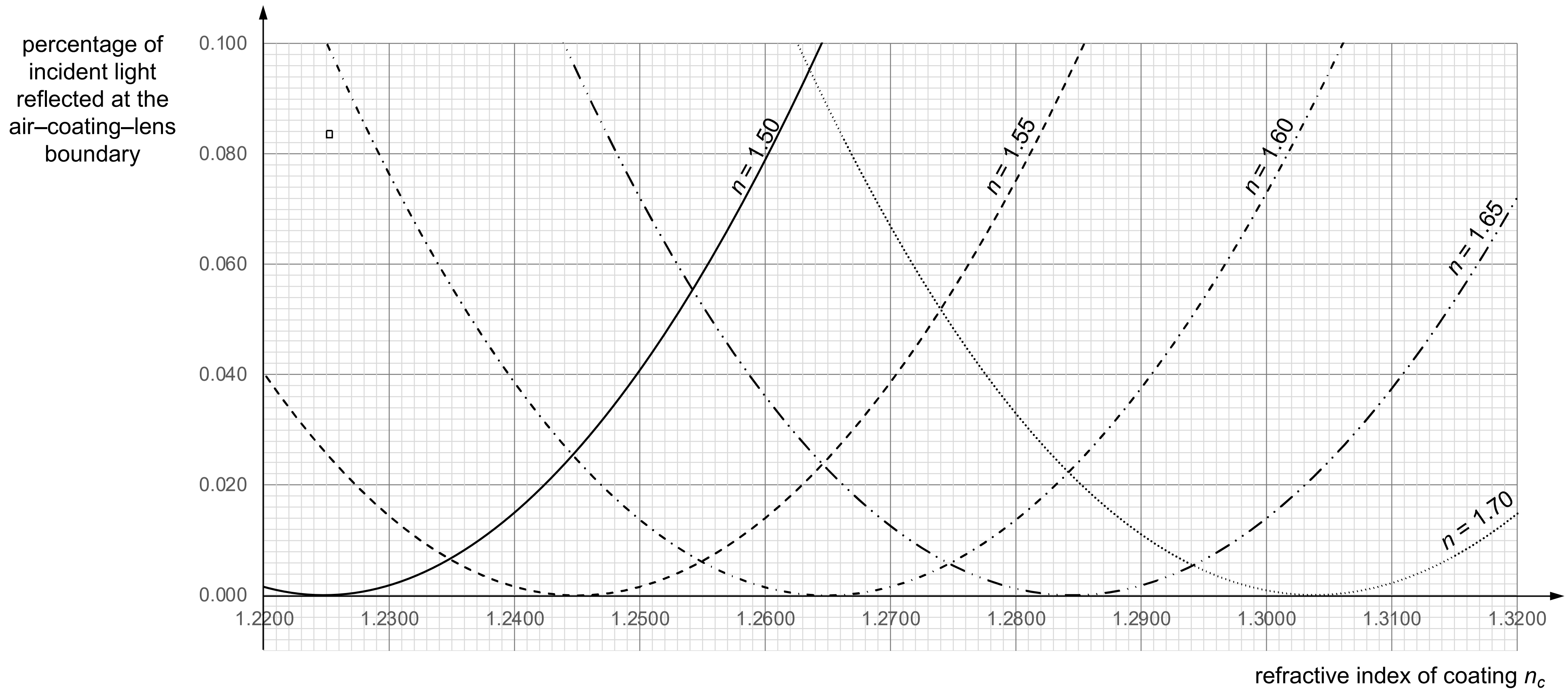


Fig. 7.3

Fig. 7.4 shows how the variation with the refractive index of the coating n_c of the percentage of light reflected at the air–coating–lens interfaces. The optimum refractive index n_c of the coating that minimises the reflection of light for different values of refractive index of lenses can obtained from Fig. 7.4.

Multiple layers of AR coatings on modern lens can virtually eliminate the reflection of visible light from eyeglass lenses, allowing 99.5 percent of available light to pass through the lenses and enter the eye for good vision.



percentage of incident light reflected at the air–coating–lens boundary
0.100
0.080
0.060
0.040
0.020
0.000
1.2200
1.2300
1.2400
1.2500
1.2600
1.2700
1.2800
1.2900
1.3000
1.3100
1.3200
refractive index of coating n_c
n = 1.50
n = 1.55
n = 1.60
n = 1.65
n = 1.70

Fig. 7.4

(a) **(i)** Show that $R = \left(\dfrac{n_2 - n_1}{n_2 + n_1}\right)^2$

[2]

(ii) **1.** Explain why the sum of R and T is one.

....

.... [1]

2. Deduce the expression for the transmittance T, in terms of n_1 and n_2.

[1]

(iii) Fig. 7.5 shows monochromatic light is incident normally on a lens without AR coating.

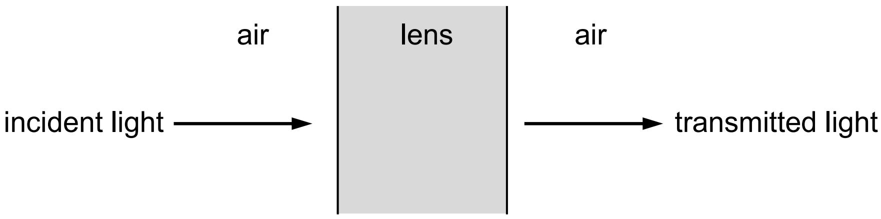


Fig. 7.5

The refractive index of air is 1.00. The refractive indices of two different lens materials are:

	CR–39	polycarbonate
refractive index	1.50	1.58

1. For CR–39 lens, show that about 92% the incident light is transmitted.

[2]

2. Determine the percentage decrease in the transmitted light for a polycarbonate lens.

percentage decrease = ………………………………… % [2]

(b) **(i)** When light wave is reflected from the surface of a medium with higher refractive index than that of the medium in which they are travelling, the reflected wave will change phase by 180° or π radians.

Explain why the reflection of a monochromatic light is minimised when the thickness of the AR coating is a quarter of the wavelength.

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

…………………………………………………………………………………………………..

………………………………………………………………………………………………… [3]

(ii) Fig. 7.6 shows just the incident wave at one instant on the lens with the AR coating.

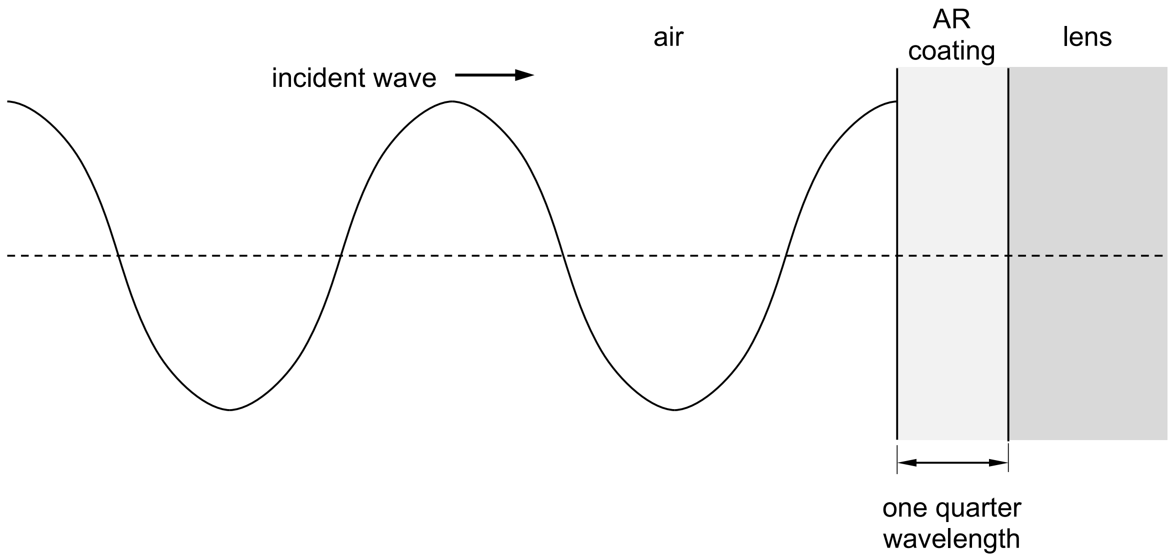


Fig. 7.6

On Fig. 7.6, sketch

1. the reflected wave from the air–coating interface and label it as “C”,

2. the reflected wave from the coating–lens interface and label it as “L”. [3]

(c) **(i)** Use data from Fig. 7.4 to complete the table below.

refractive index n of lens	optimum refractive index n_c of the coating
1.50	1.2250
1.55	
1.60	
1.65	
1.70	

[1]

(ii) Without drawing a graph, use your data in **(c)(i)** to verify that n_c is proportional to the square root of n.

[2]

(iii) Determine the optimum refractive index of the coating for polycarbonate lens.

optimum refractive index = ... [1]

(d) **(i)** Suggest why there are multiple layers of AR coatings on modern lens.

....

....

.... [2]

(ii) Some high-quality lenses have as many as seven layers of AR coatings.

Estimate the thickness of the AR coatings.

thickness = ... m [2]

[Total: 22]

[Turn over

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NATIONAL JUNIOR COLLEGE

SENIOR HIGH 2 PRELIMINARY EXAM

Higher 2

CANDIDATE NAME

SUBJECT CLASS

REGISTRATION NUMBER

PHYSICS
Paper 3 Longer Structured Questions
Candidate answers on the Question Paper.

No Additional Materials are required.

9749/03

17 Sep 2020
2 hours

READ THESE INSTRUCTION FIRST
Write your subject class, registration number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

The use of an approved scientific calculator is expected, where appropriate.

Section A
Answers **all** questions.

Section B
Answer **one** question only.

You are advised to spend one and a half hours on Section A and half an hour on Section B.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	**/ 7**
2	**/ 9**
3	**/ 6**
4	**/ 8**
5	**/ 12**
6	**/ 8**
7	**/ 8**
8	**/ 2**
Section B	
9	**/ 20**
10	**/ 20**
Total (80m)	

This document contains **28** printed pages and **0** blank pages.

Data

speed of light in free space	$c = 3.00 \times 10^{8}\,\mathrm{m\,s^{-1}}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
	$(1/(36\pi)) \times 10^{-9}\,\mathrm{F\,m^{-1}}$
elementary charge	$e = 1.60 \times 10^{-19}\,\mathrm{C}$
the Planck constant	$h = 6.63 \times 10^{-34}\,\mathrm{J\,s}$
unified atomic mass constant	$u = 1.66 \times 10^{-27}\,\mathrm{kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31}\,\mathrm{kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27}\,\mathrm{kg}$
molar gas constant	$R = 8.31\,\mathrm{J\,K^{-1}\,mol^{-1}}$
the Avogadro constant	$N_A = 6.02 \times 10^{23}\,\mathrm{mol^{-1}}$
the Boltzmann constant	$k = 1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
acceleration of free fall	$g = 9.81\,\mathrm{m\,s^{-2}}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
work done on / by a gas	$W = p\Delta V$
hydrostatic pressure	$p = \rho gh$
gravitational potential	$\phi = -Gm/r$
temperature	$T/\mathrm{K} = T/{}^\circ\mathrm{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V} < c^2 >$
mean translational kinetic energy of an ideal gas molecule	$E = \frac{3}{2}kT$
displacement of particle in s.h.m.	$x = x_0 \sin\omega t$
velocity of particle in s.h.m.	$v = v_0 \cos\omega t$ $= \pm\,\omega\sqrt{x_0^{\,2} - x^2}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$1/R = 1/R_1 + 1/R_2 + \ldots$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	$x = x_0 \sin\omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 nI$
radioactive decay	$x = x_0 \exp(-\lambda t)$
decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Section A

Answer **all** the questions in the spaces provided.

1 Fig. 1.1. shows an airplane dropping a crate of emergency supplies.

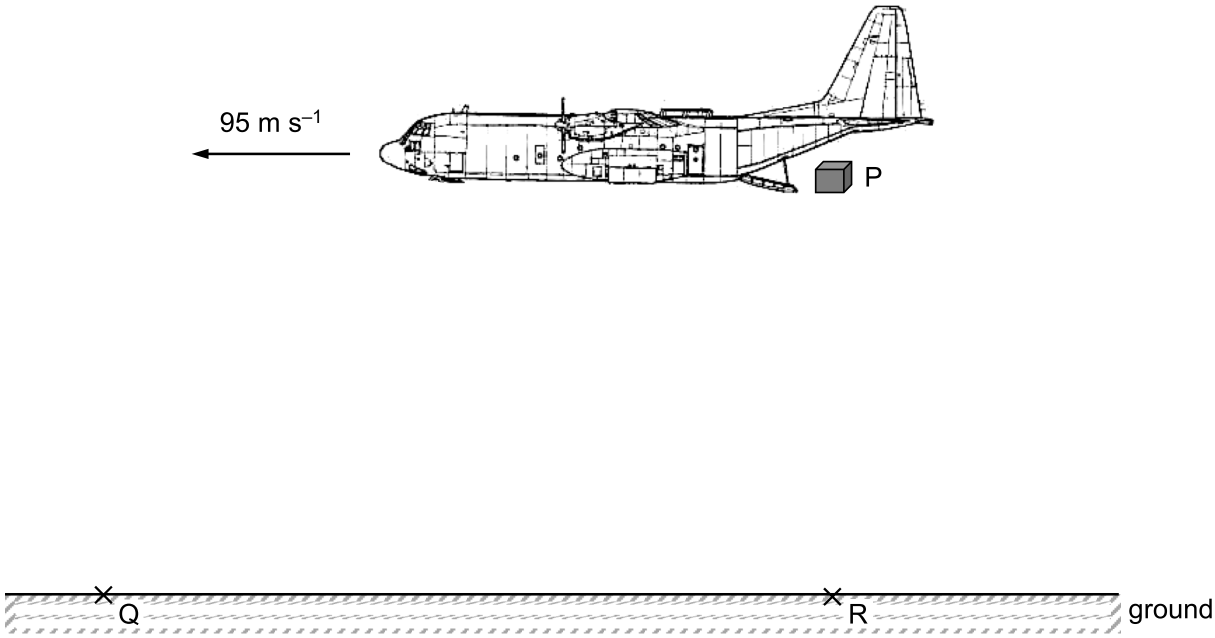


Fig. 1.1

The airplane is travelling horizontally at a constant speed 95 m s^{-1}. The crate is released at point P and it lands at point Q.

The air resistance acting on the crate may be neglected.

(a) On Fig. 1.1, sketch the path of the crate. [1]

(b) Explain why the horizontal component of the crate's velocity is constant when it is moving between point P and Q.

………

……… [1]

(c) The maximum vertical component of the crate's velocity just before landing is 32 m s^{-1} to prevent damaging the supplies.

(i) Show that the maximum height of point P is 52 m.

[1]

(ii) Point R is vertically below point P.

Calculate the horizontal distance QR.

QR = ………………………………………. m [2]

(d) Suggest and explain the new landing point relative to Q if air resistance is not negligible.

……

……

……

……………………………………………………………………………………………….. [2]

[Total: 7]

[Turn over

2 **(a)** The gravitational potential ϕ of a point at a distance r from an isolated point mass m is given by the expression

$$\phi = -\frac{Gm}{r}$$

where G is the gravitational constant.

Explain why gravitational potential is negative.

....

....

.... [2]

(b) A space probe of mass m is launched from the geographical North Pole, as shown in Fig. 2.1.

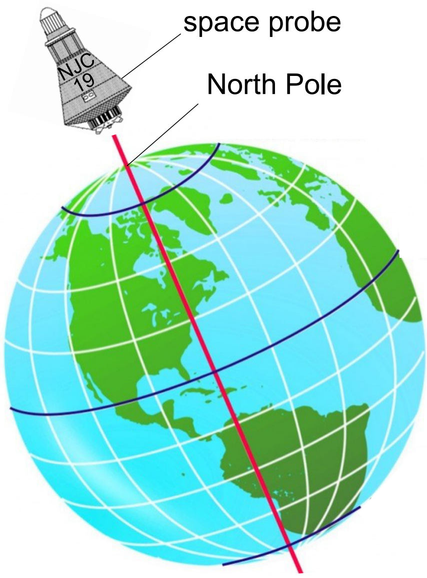


Fig. 2.1

The probe is provided with energy equivalent to an initial kinetic energy E during the launch given by the expression

$$E = \frac{3GMm}{4R_E}$$

where M is the mass of the Earth and R_E is the radius of the Earth.
Work done in overcoming resistive forces in Earth's atmosphere can be neglected.

(i) Explain why the probe cannot reach infinity with this energy E.

....

....

....

....

.... [2]

(ii) The mass m of the probe is 1000 kg, the mass M of Earth is 6.0×10^{24} kg and the radius R_E of the Earth is 6.4×10^6 m.

Calculate the distance from the centre of Earth at which the probe will come to a rest momentarily.

distance = ……………………………………… m [3]

(iii) Explain why the distance calculated in **(b)(ii)** is further if the space probe is launched from the equator in the eastward direction.

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

……………………………………………………………………………………………………..

…………………………………………………………………………………………………… [2]

[Total: 9]

3 To celebrate the Millennium in the year 2000, a footbridge was constructed across the River Thames in London. After the bridge was opened to the public it was discovered that the structure could easily be set into oscillation when large numbers of pedestrians were walking across it.

(i) State the name of the phenomenon described above.

.... [1]

(ii) Explain why this phenomenon can become hazardous.

....

....

....

....

....

.... [3]

(iii) Engineers looked at the situation and applied more damping to the bridge.

Fig. 3.1 shows the variation with driving frequency of the amplitude of the oscillation of the bridge after applying the damping.

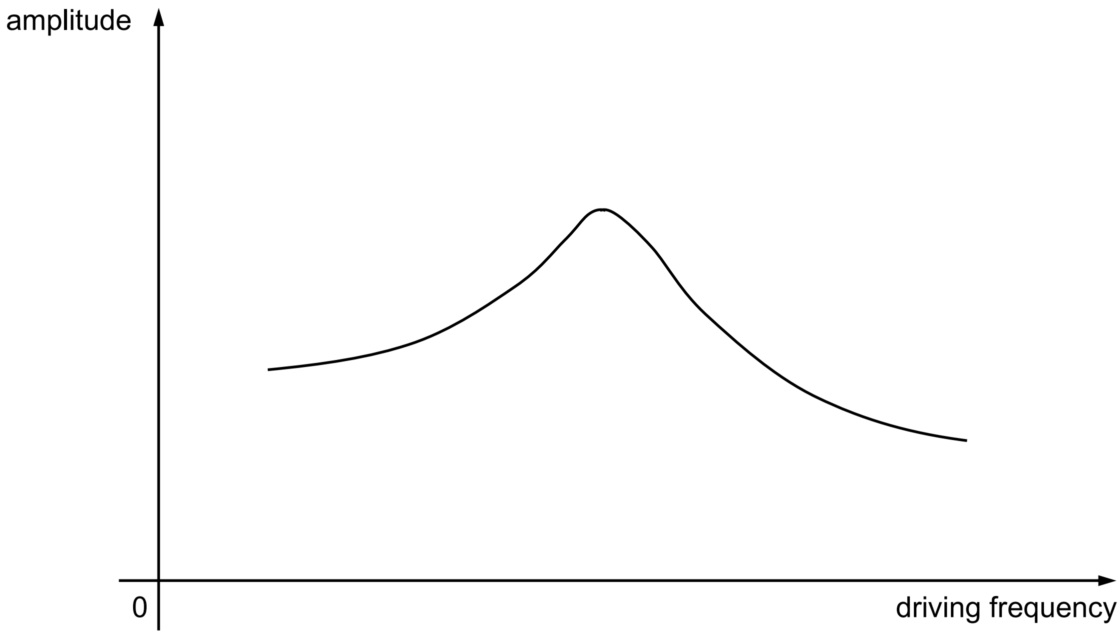


Fig. 3.1

On Fig. 3.1, sketch the graph of the variation with driving frequency of the amplitude before damping was applied. [2]

[Total: 6]

Question 4 begins over the page.

[Turn over

4 A progressive wave from a source moves past two points P and Q.

Fig. 4.1 shows the variation with time t of the displacement y at P.

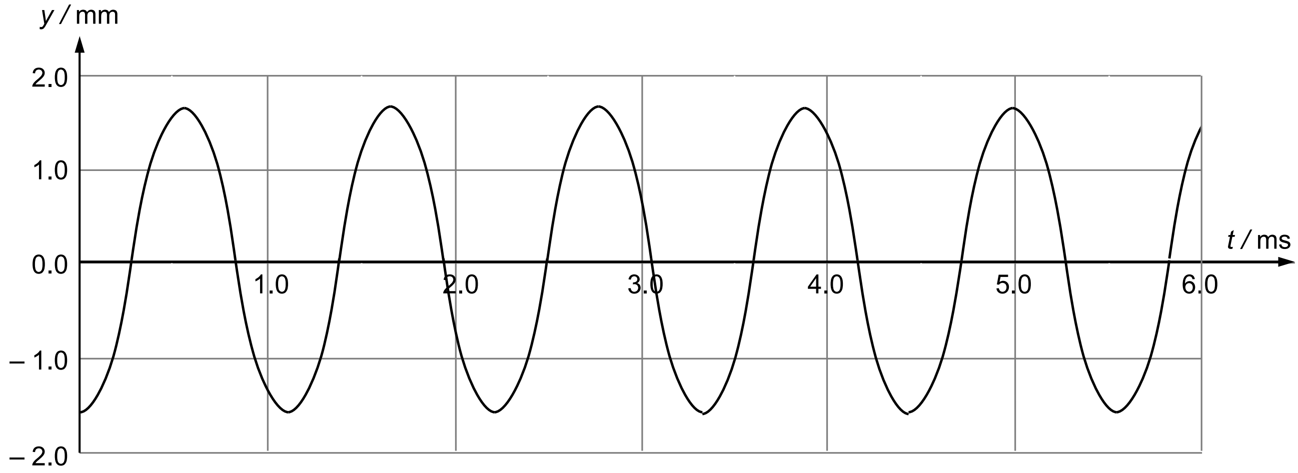


Fig. 4.1

Fig. 4.2 shows how the displacement of the wave at time t = 0 varies with distance x from the source.

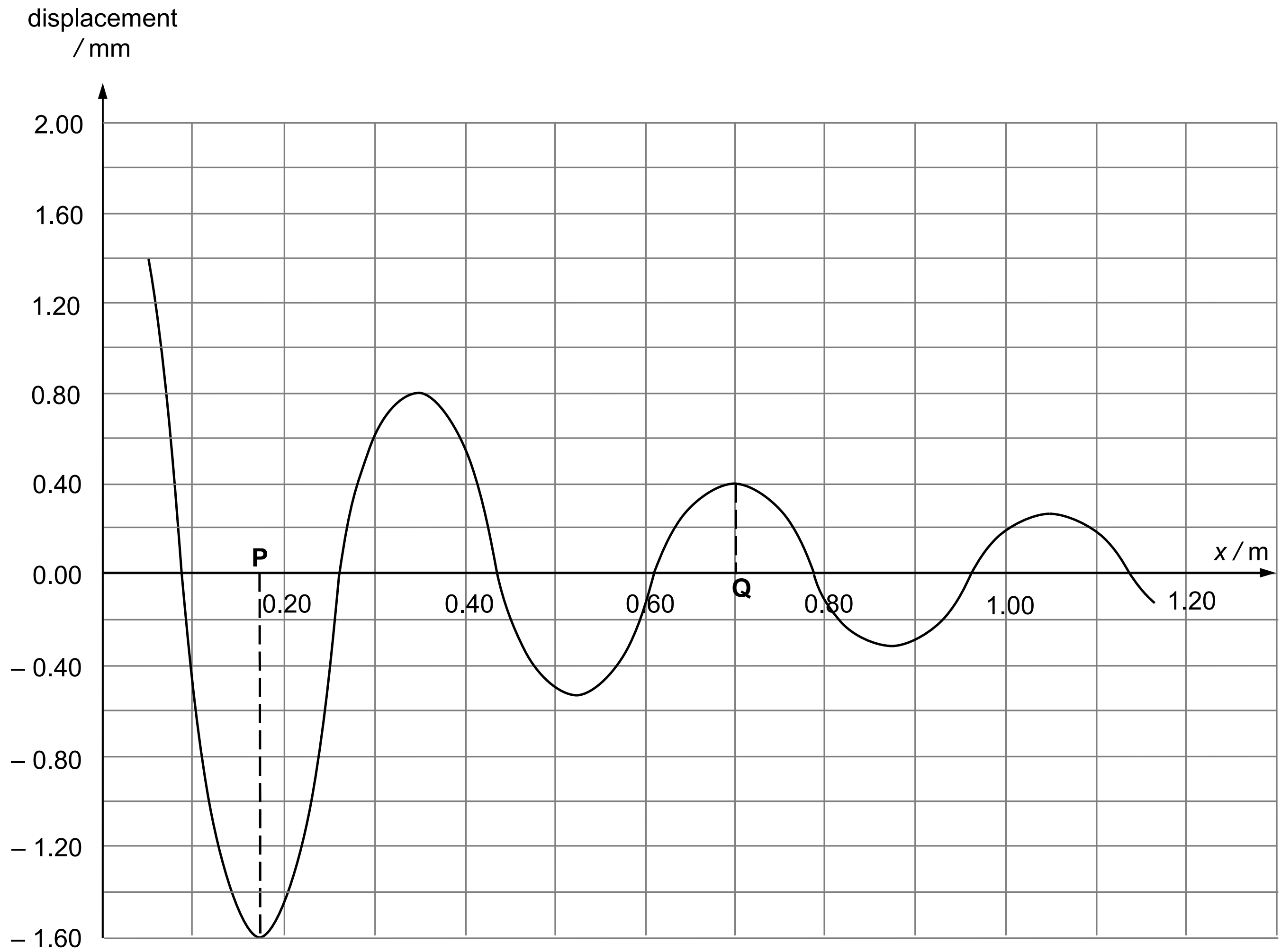


Fig. 4.2

(a) Explain what is meant by the term *progressive* as applied to a wave.

……

……

……[2]

(b) Using data from Fig. 4.1 and 4.2, deduce for this wave

(i) the wavelength,

wavelength = ……………………………………… m [1]

(ii) the frequency,

frequency = …………………………………….. Hz [1]

(iii) the speed.

speed = ………………………………….. m s^{-1} [1]

(c) The intensity of the wave is inversely proportional to the square of the distance from the source.

Without drawing a graph, use the data from Fig. 4.1 and 4.2 to determine whether the wave source in **(b)** obeys this relationship.

conclusion ………………………………………………………………………………………[3]

[Total: 8]

5 **(a)** The resistance R of a metal is given by

$$R = \rho \frac{L}{A}$$

where ρ is the resistivity of the material of a wire, L is the length of the wire and A is the circular cross-sectional area of the wire.

An experiment is performed to determine the resistivity ρ of the wire. The average of the measurements, with their actual uncertainties, are :

diameter of wire / mm	length of wire, L / cm	resistance of wire, R / Ω
0.23 ± 0.01	40.0 ± 0.1	4.80 ± 0.05

Calculate the actual uncertainty in ρ .

actual uncertainty in ρ =……………………….. Ω m [3]

5 **(b)** A battery of electromotive force (e.m.f.) 6.0 V and internal resistance *r* is connected to an external resistor of resistance 12 Ω and a NTC thermistor X, as shown in Fig. 5.1.

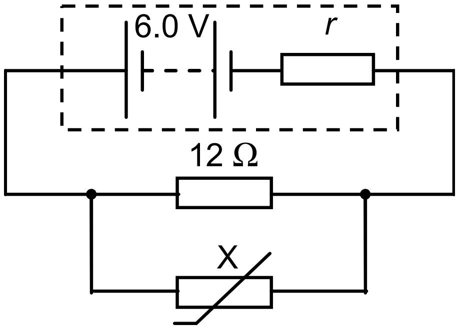


Fig. 5.1

(i) Explain, in terms of energy changes in the circuit, why the potential difference across the terminals of the battery is less than the e.m.f.

....

....

.... [2]

(ii) The combined resistance of the 12 Ω resistor and thermistor X connected in parallel is 4.8 Ω.

Determine the ratio

$$\frac{\text{power dissipated in thermister X}}{\text{power dissipated in 12 }\Omega\text{ resistor}}.$$

ratio = ... [3]

(iii) The temperature of NTC thermistor X is now decreased.

State and explain the effect, if any, of this temperature change on the total power produced by the battery.

....

....

.... [2]

[Turn over

(c) The I–V characteristic of a 60 W filament lamp is shown in the fig. 5.2.

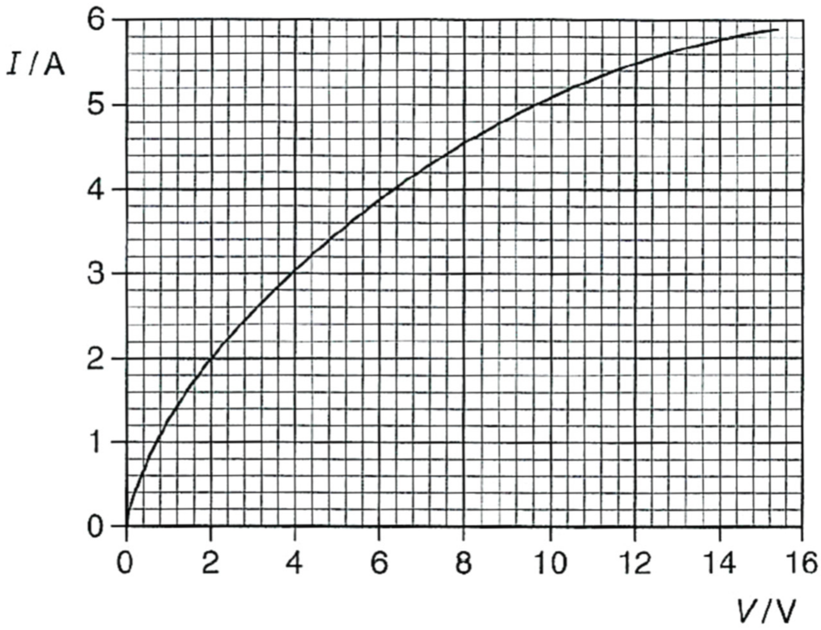


Fig. 5.2

(i) Suggest why the graph stops at 15.4 V.

....

.... [1]

(ii) Explain why the resistance of the lamp is **not** the reciprocal of the gradient of the graph.

....

....

.... [1]

[Total: 12]

Question 6 begins over the page.

6 **(a)** Define *magnetic flux*.

....

.... [1]

(b) A rectangular coil has dimension 20 mm by 35 mm and 650 turns. The coil rotates about a horizontal axis perpendicular to a uniform magnetic field of flux density 2.5×10^{-3} T, as shown in Fig. 6.1.

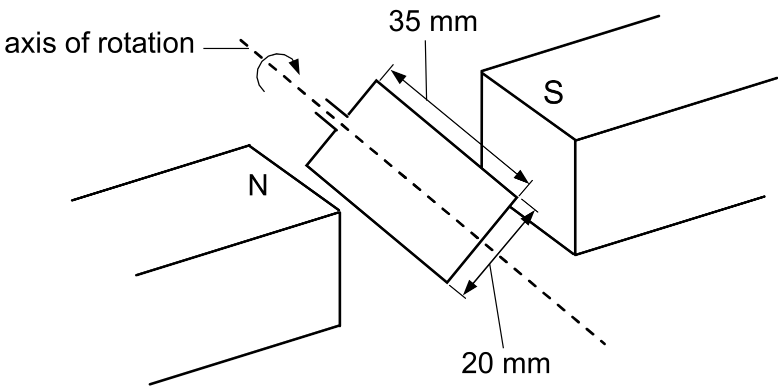
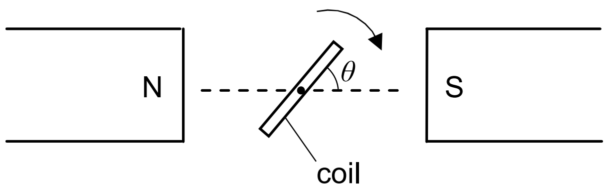


Fig. 6.1

The plane of the coil makes an angle θ with the horizontal, as shown in Fig. 6.1.



(front view)

Fig. 6.1

The coil turns from $\theta = 30°$ to $\theta = 0°$ in 5.0 ms.

The coil has a resistance of 5.0 Ω.

(i) Calculate the magnetic flux linkage of the coil when $\theta = 30°$. Give an appropriate unit.

magnetic flux linkage = ... [3]

(ii) Calculate the average induced e.m.f. across the coil.

e.m.f. = ... V [2]

(iii) Calculate the amount of charge flow through the coil in 5.0 ms.

charge = ... C [2]

[Total: 8]

7 **(a)** Coil X of 1200 turns of insulated wire is wound on to a soft-iron core.

The coil is connected in series with a battery, a switch and an ammeter, as shown in Fig. 7.1.

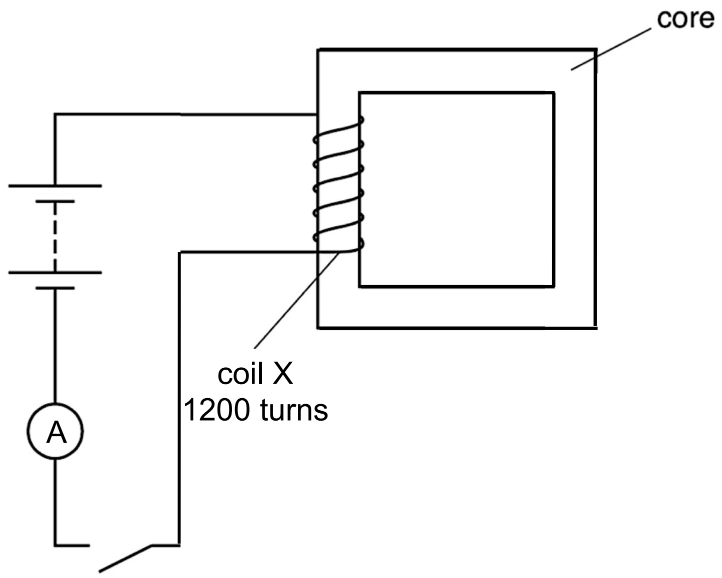


Fig. 7.1

Use the laws of electromagnetic induction to explain why, when the switch is closed, the current increases **gradually** to its maximum value.

....

....

....

....

....

....[3]

(b) Coil Y is now wound on the soft iron, as shown in Fig. 7.2.

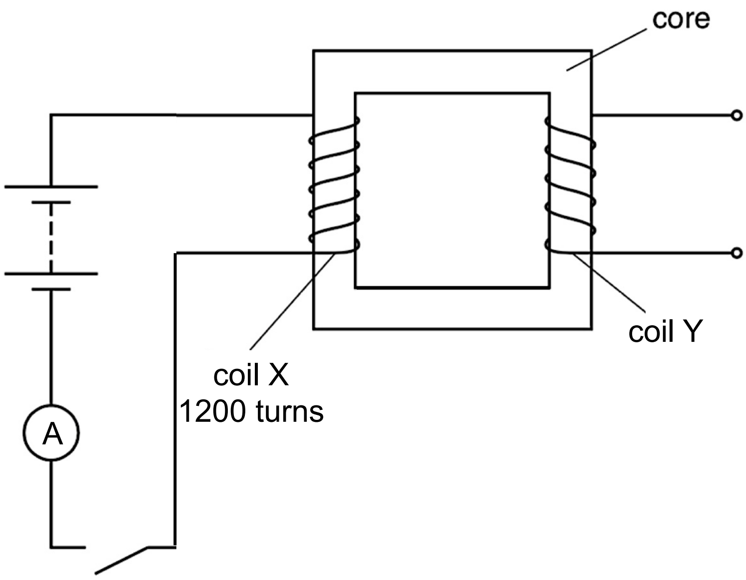


Fig. 7.2

Explain why, when the switch is closed and maximum current is reached in coil X, an electromotive force (e.m.f.) is not induced at coil Y.

....

....

....

....[2]

(c) The battery in **(b)** is now replaced by an alternating voltage of r.m.s. value 90 V. The variation with time t of the e.m.f. E induced across the secondary coil is shown in Fig. 7.3.

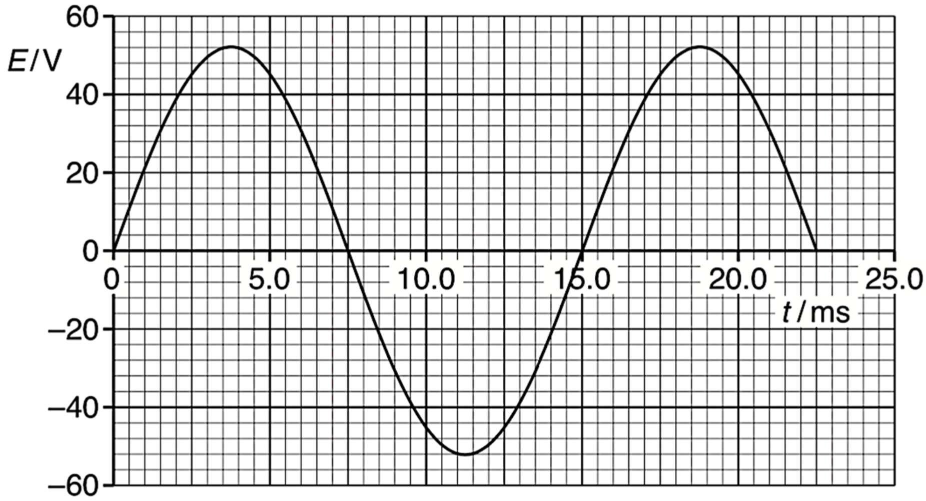


Fig. 7.3

Use data from Fig. 7.3 to

(i) calculate the number of turns of the secondary coil,

number = ... [2]

(ii) state a time when the magnetic flux linking the secondary coil is a maximum.

time = ... ms [1]

[Total: 8]

8 The radioactive decay of the isotope strontium-90 is both spontaneous and random.

Explain what is meant by

(a) *spontaneous decay,*

....

.... [1]

(b) *random decay*.

....

....

....

.... [1]

[Total: 2]

Section B

Answer **one** question from this Section in the spaces provided.

9 **(a)** **(i)** Explain what is meant by the *internal energy of an ideal gas*.

……

……………………………………………………………………………………………………….. [1]

(ii) A sealed container of volume V contains ideal gas at pressure p.

Derive an expression for the internal energy of the gas in terms of p and V.

[2]

(b) **(i)** The mass of one mole of carbon dioxide is 44.0 g. Show that the mass of one molecule of carbon dioxide is about 7.31×10^{-26} kg.

[1]

(ii) The surface temperature of the planet Mercury is 467 °C.

Determine the root-mean-square speed of one molecule of carbon dioxide at this temperature.

root-mean-square speed = ………………………………….. m s^{-1} [3]

(iii) The escape speed of Mercury is 4250 m s^{-1}.

Suggest, with a reason, whether an atmosphere of carbon dioxide can exist on the surface of Mercury.

……

……

……………………………………………………………………………………………………….. [2]

(c) **(i)** State the first law of thermodynamics.

…………………………………………………………………………………………………...

…………………………………………………………………………………………………...

………………………………………………………………………………………………….. [2]

(ii) A fixed mass of ideal gas undergoes the cycle ABCDA, as shown in Fig. 9.1.

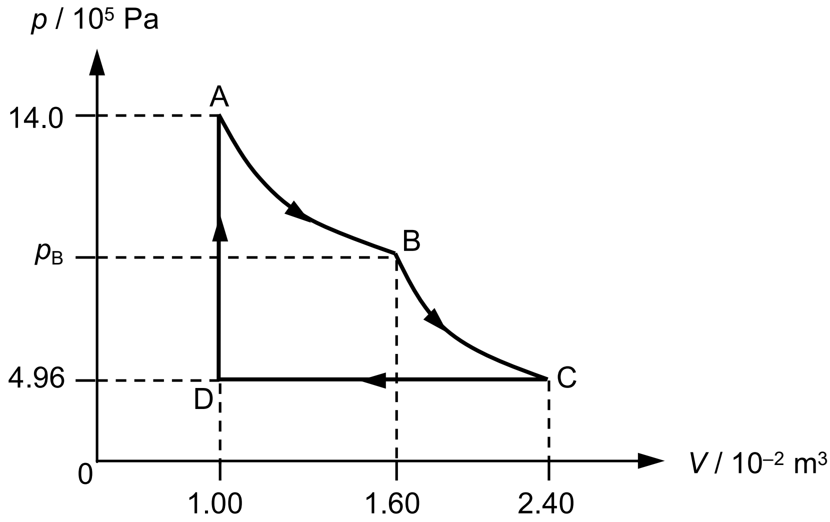


Fig. 9.1

The gas expands from A to B isothermally. It undergoes further expansion from B to C without any exchange of thermal energy.
The gas is compressed at constant pressure from C to D.
From D to A, the pressure of the gas is increased at constant volume.

1. Show that the pressure of the gas at point B is 8.75×10^5 Pa.

[1]

2. Some energy changes for the cycle in Fig. 9.1 are shown in Fig. 9.2.

	heat supplied to gas / J	work done on gas / J	increase in internal energy of gas / J
A to B		– 6580	
B to C			– 3140
C to D			– 10400

Fig. 9.2

Complete Fig. 9.2. [3]

[Turn over

3. Calculate the increase in the internal energy of the gas from D to A.

increase = ………………………………………. J [2]

(iii) The ideal gas in part **(ii)** is transferred into a container with a movable frictionless piston of small mass, as shown in the Fig. 9.3.

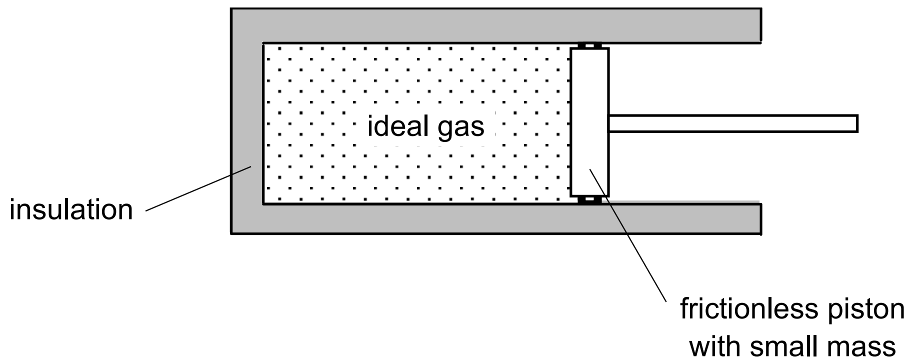


Fig. 9.3

The container and the piston are perfect thermal insulators.

Explain, in terms of the motion of the ideal gas molecules, why the temperature of the gas decreases when the gas expands.

……………………………………………………………………………………………………...

……………………………………………………………………………………………………...

……………………………………………………………………………………………………...

……………………………………………………………………………………………………...

……………………………………………………………………………………………………...

…………………………………………………………………………………………………….. [3]

[Total: 20]

Question 10 begins over the page.

10 **(a)** Explain why, for the photoelectric effect, the existence of a threshold frequency provides evidence for the particulate nature of electromagnetic radiation, as opposed to a wave theory.

....

....

....

....

....

....

....

.... [4]

(b) An electron is accelerated from rest in a vacuum through a potential difference of 4.7 kV.

(i) Calculate the de Broglie wavelength of the accelerated electron.

wavelength = ... m [4]

(ii) By reference to your answer in **(b)(i)** suggest why such electrons may assist with an understanding of the crystal structure.

....

....

....

.... [2]

(c) The emission spectrum of isolated hydrogen atoms consists of a number of discrete wavelengths.

(i) Explain how the spectrum of hydrogen provides evidence for the existence of discrete energy levels in atoms.

....

....

....

....

....

.... [3]

(ii) Some electron energy levels of the hydrogen atom are illustrated in Fig. 10.1.

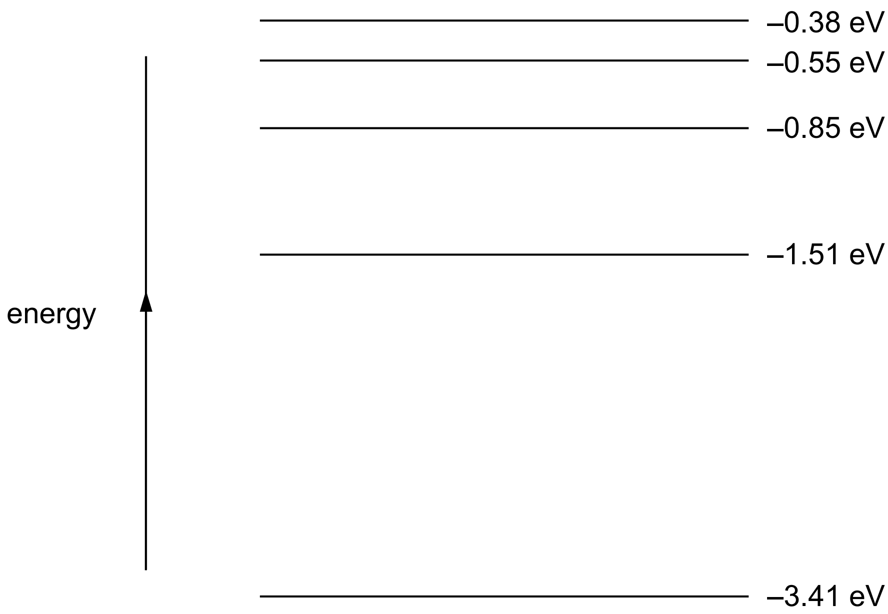


Fig. 10.1 (not to scale)

The minimum energy of a photon of visible light is 1.78 eV.

1. On Fig. 10.1, draw the transition giving rise to the longest wavelength in the visible spectrum. [1]

2. Use Fig. 10.1 to determine the shortest wavelength of the photons emitted.

wavelength = ...m [2]

(iii) The light in a beam has a continuous spectrum of wavelengths from 400 nm to 700 nm. The light is incident on some cool hydrogen gas, as illustrated in Fig. 10.2.

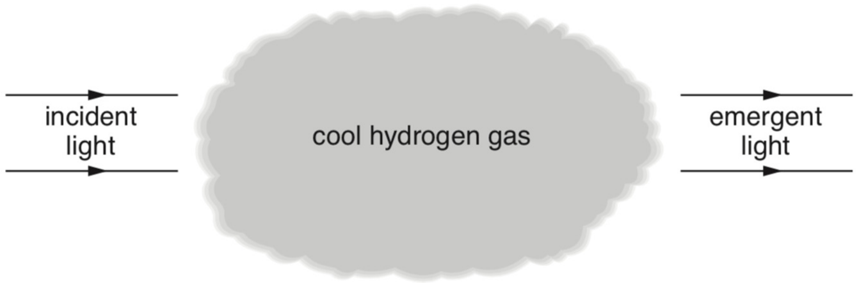


Fig. 10.2

Using the energy levels of the hydrogen atom in Fig. 10.1, state and explain the appearance of the spectrum of the emergent light.

....

....

....

....

....

....

....

.... [4]

[Total: 20]

2020 H2 Physics Paper 1 Debrief

Question Number	*Key*	*Question Number*	*Key*
1	**D**	16	**C**
2	**A**	17	**A**
3	**B**	18	**C**
4	**D**	19	**A**
5	**D**	20	**A**
6	**A**	21	**D**
7	**B**	22	**C**
8	**D**	23	**A**
9	**C**	24	**B**
10	**A**	25	**A**
11	**D**	26	**B**
12	**C**	27	**A**
13	**C**	28	**C**
14	**A**	29	**D**
15	**D**	30	**A**

1. The speed of a wave in deep water depends on its wavelength L and the acceleration of free fall g.

What is a possible equation for the speed v of the wave?

A $v = \dfrac{2\pi g}{L}$ **B** $v = \dfrac{gL}{4\pi^2}$ **C** $v = 2\pi\sqrt{\dfrac{g}{L}}$ **D** $v = \sqrt{\dfrac{gL}{2\pi}}$

Answer: D

Units for the RHS equals to units of v (m s^{-1})

A: $\dfrac{\text{m s}^{-2}}{\text{m}} = \text{s}^{-2}$

B: $\text{m s}^{-2} \times \text{m} = \text{m}^2\ \text{s}^{-2}$

C: $\left(\dfrac{\text{m s}^{-2}}{\text{m}}\right)^{1/2} = \text{s}^{-1}$

D: $\left(\text{m s}^{-2} \times \text{m}\right)^{1/2} = \text{m s}^{-1}$

2. The power P dissipated in a resistor of resistance R is calculated using the expression

$$P = \frac{V^2}{R}$$

where V is the potential difference (p.d.) across the resistor.

In an experiment to determine P, R is measured to within a percentage uncertainty of 2%.

What is the maximum percentage uncertainty in the measurement of V such that P can be determined with a maximum uncertainty of ± 5%?

A 1.5% **B** 3.0% **C** 3.5% **D** 7%

Answer: A

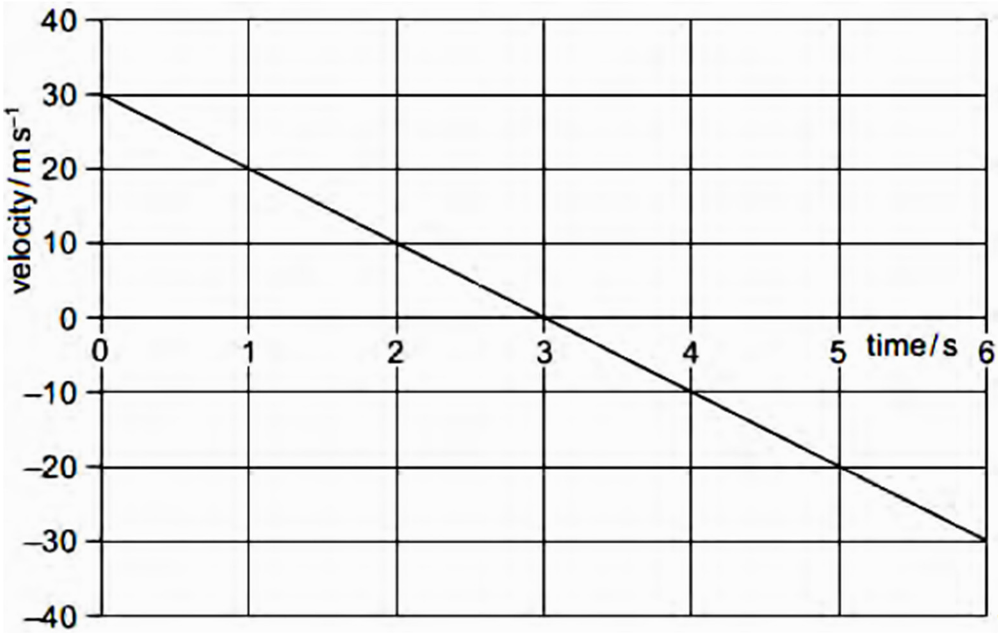
$$P = \frac{V^2}{R}$$

$$\Rightarrow \frac{\Delta P}{P}\frac{\Delta V}{V} = 2\frac{\Delta V}{V} + \frac{\Delta R}{R}$$

$$0.05 = 2\frac{\Delta V}{V} + 0.02$$

$$\frac{\Delta V}{V} = 0.015$$

3. A stone is thrown vertically upwards. A student plots the variation with time of its velocity.



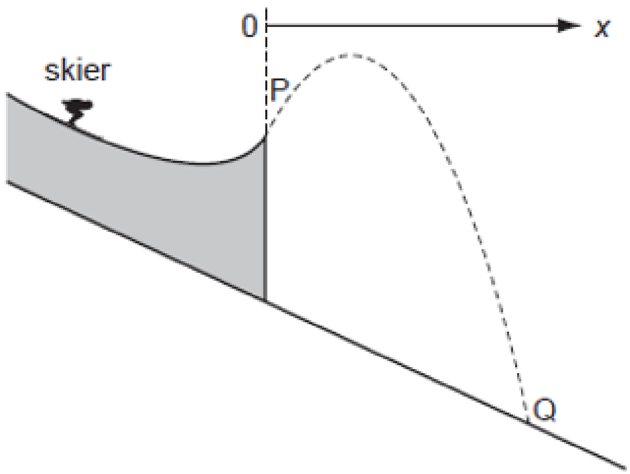
What is the vertical displacement of the stone from its starting point after 5 seconds?

A 20 m **B** 25 m **C** 45 m **D** 65 m

Answer: B

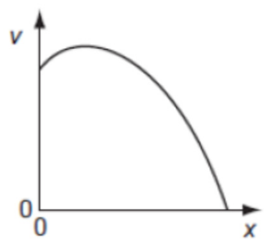
distance = area under graph $= \frac{1}{2} \times 30 \times 3 - \frac{1}{2} \times 20 \times 2 = 25$ m

4. The dotted line shows the path of a competitor in a ski-jumping competition.

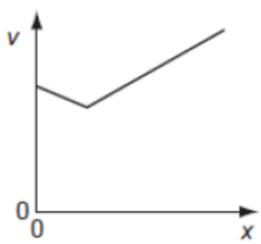


Ignoring air resistance, which graph best represents the variation of his speed v with the horizontal distance x covered from the start of his jump at P before landing at Q?

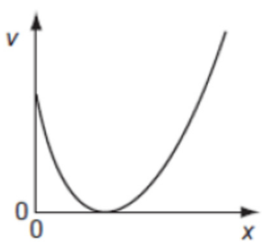
A



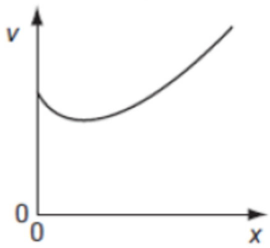
B



C



D



Answer: D

- speed v decreases from P to highest point and increases from highest point
- v cannot be zero due to horizontal velocity is constant
  At this point, you may narrow your options to B and D

<u>Quantitative</u>
Let u_x be horizontal speed and u_y be vertical speed at P. and v be speed between P and Q.
Let h be the vertical displacement from P.
Using energy conservation,

$$\Delta E_K + \Delta E_P = 0 \Rightarrow \tfrac{1}{2}mv^2 - \tfrac{1}{2}mu^2 + mgh = 0$$

$$v^2 = u^2 - 2gh \qquad \text{...... (1)}$$

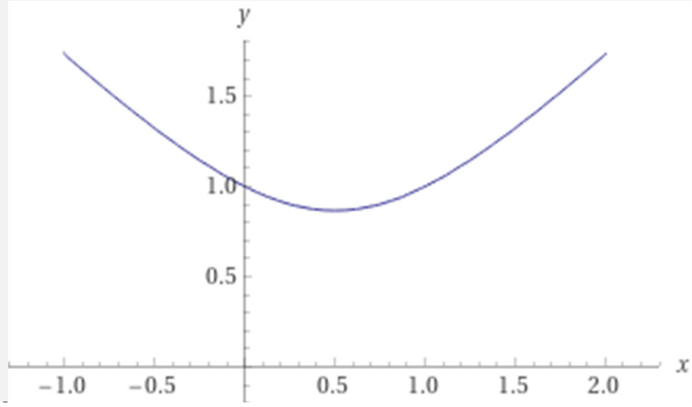
vertically: $$h = u_y t - \tfrac{1}{2}gt^2 \qquad \text{...... (2)}$$

horizontally: $$x = u_x t \Rightarrow t = \frac{x}{u_x} \qquad \text{...... (3)}$$

sub (3) into (2): $$h = \frac{u_y}{u_x}x - \frac{g}{2u_x^{\,2}}x^2 \qquad \text{...... (4)}$$

sub (4) into (1): $$v^2 = u^2 - \frac{2gu_y}{u_x}x + \frac{g^2}{u_x^{\,2}}x^2$$

You can use graphing calculators to plot $y^2 = 1 - x - x^2$



5. Two magnets P and Q, having masses M_P and M_Q respectively, exert forces on each other and have no other external forces acting on them. The force acting on P is F, which gives P an acceleration a.

Which of the following pairs is correct?

	Magnitude of force on Q	Magnitude of acceleration of Q
A	$\frac{M_Q}{M_P}F$	a
B	$\frac{M_P}{M_Q}F$	a
C	F	$\frac{M_Q}{M_P}a$
D	F	$\frac{M_P}{M_Q}a$

Answer: D

By Newton's 3rd Law, force on P = force on Q

$F = M_P a$ …… (1)

$F = M_Q a_Q$ …… (2)

6. Particles X (of mass 4 units) and Y (of mass 9 units) move directly towards each other, collide and then separate.

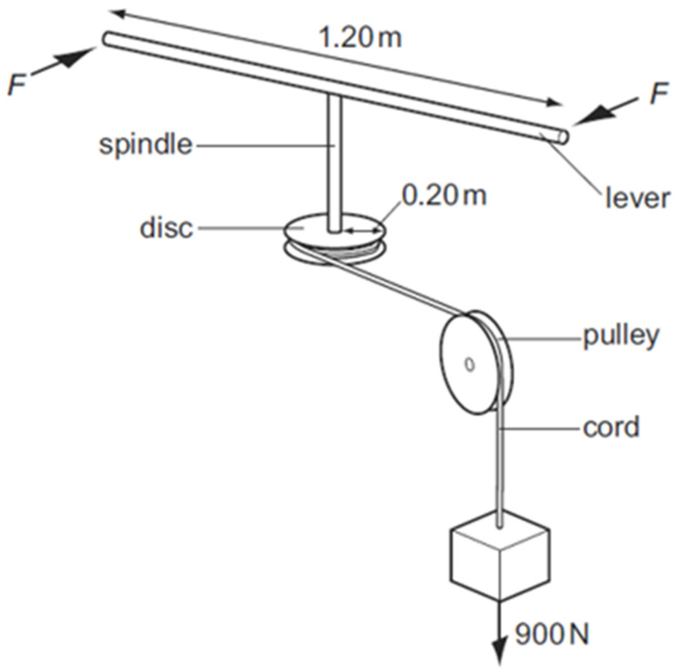
If Δv_X is the change in velocity of X and Δv_y is the change in velocity of Y, the magnitude of ratio $\frac{\Delta v_X}{\Delta v_y}$ is

A 9 / 4 **B** 3 / 2 **C** 1 **D** 2 / 3

Answer: A

magnitude of change of momentum of X and Y equal, so $m_x \Delta v_x = m_y \Delta v_y \Rightarrow \frac{\Delta v_x}{\Delta v_y} = \frac{m_y}{m_x} = \frac{9}{4}$

7. A spindle is attached at one end to the centre of a lever 1.20 m long and at its other end to the centre of a disc of radius 0.20 m. A cord is wrapped round the disc, passes over a pulley and is attached to a 900 N weight.



What is the minimum force *F*, applied to each end of the lever, that could lift the weight?

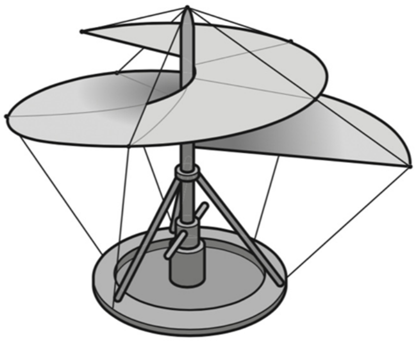
A 75 N **B** 150 N **C** 300 N **D** 950 N

Answer: B

torque couple on lever $= F \times 1.20$
moment on disc $= 900 \times 0.20$

To lift weight, $F \times 1.20 > 900 \times 0.2 \Rightarrow F > 150$ N

8. Leonardo da Vinci proposed a flying machine that would work like a screw to lift the pilot into the air. The 'screw' is rotated by the pilot.



The machine and the pilot together have a total mass of 120 kg.

Which useful output power must the pilot provide to move vertically upwards at a constant speed of 2.5 m s^{-1}?

A	48 W	**B**	300 W	**C**	470 W	**D**	2900 W

Answer: D

[Method 1]

upward force provided by pilot to move at constant speed = weight

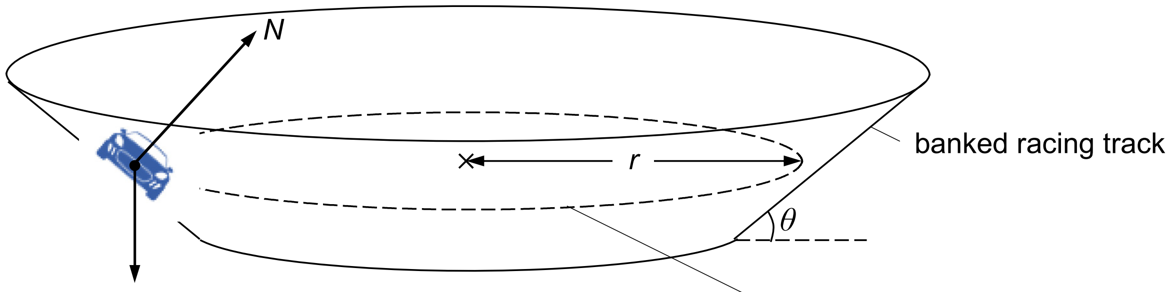
$$\text{power} = \text{force} \times \text{velocity} = 120 \times 9.81 \times 2.5 = 2943 \text{ W}$$

[Method 2]

$$\text{gain in g.p.e.} = mgv \times \text{time}$$

$$\text{power supplied} = \frac{\text{gain in g.p.e.}}{\text{time}} = mgv = 120 \times 9.81 \times 2.5 = 2943 \text{ W}$$

9. A racing car of mass m is moving with speed v on a banked racing track. The angle of inclination of the track is θ. The centre of gravity (c.g.) of the car moves along a circular path of horizontal radius r.



The centripetal force arises entirely from a component of the normal reaction N from the track.

Which of the following relations is correct?

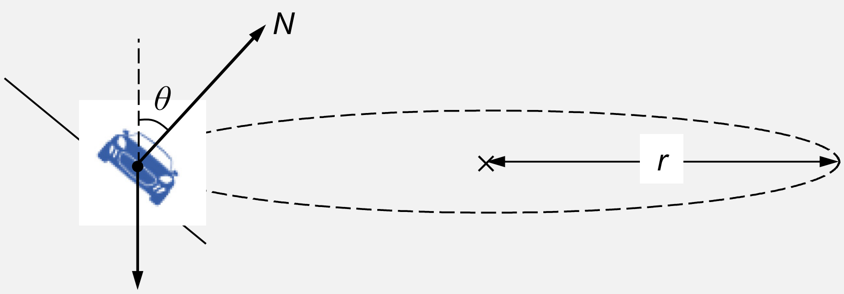
A $N \sin\theta = mg$

B $N = mg\cos\theta$

C $rg\tan\theta = v^2$

D $v^2 \tan\theta = rg$

Answer: C



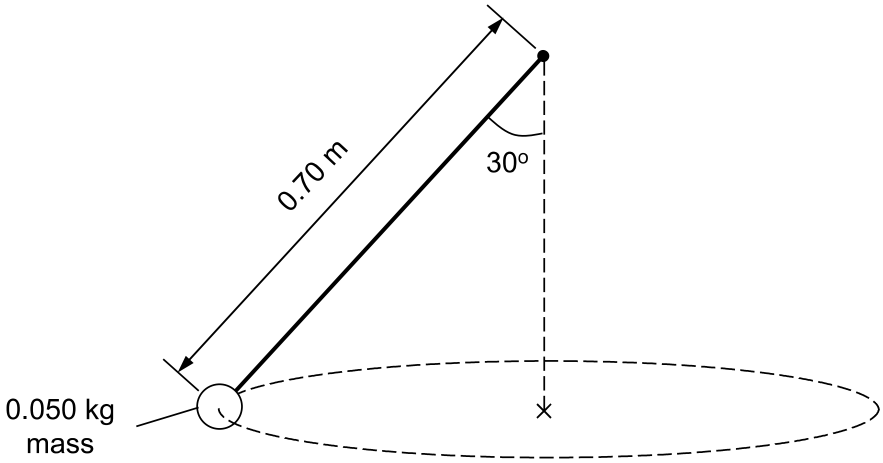
horizontal direction: $N\sin\theta = \dfrac{mv^2}{r}$ …… (1)

vertical direction: $N\cos\theta = mg$ …… (2)

(1) / (2): $\tan\theta = \dfrac{v^2}{rg}$

10. A mass of 0.050 kg is attached to one end of a piece of elastic cord of unstretched length 0.50 m. The elastic cord obeys Hooke's Law and its force constant is 40 N m^{-1}.

The mass is rotated in a horizontal circular path as shown in the diagram.



The length of the cord is 0.70 m.

What is the speed of the mass?

A 5.3 m s^{-1} **B** 7.5 m s^{-1} **C** 9.2 m s^{-1} **D** 28 m s^{-1}

Answer: A

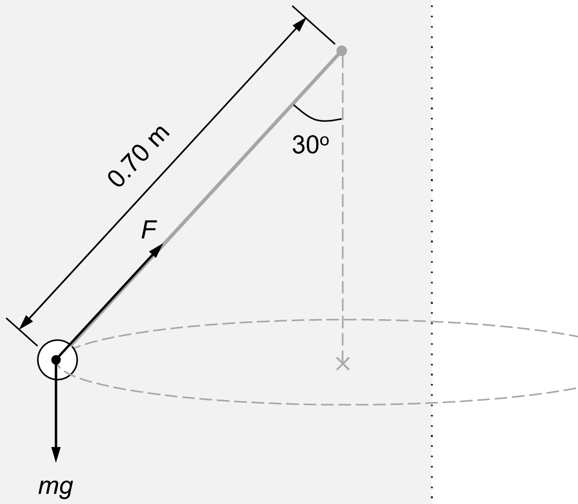
force on mass by cord $= 40 \times (0.70 - 0.50) = 8$ N

radius of circular path $= 0.70 \sin 30^\circ$

since horizontal component of force by cord provides centripetal force,

$$8 \sin 30^\circ = \frac{0.050 v^2}{0.7 \sin 30^\circ}$$

$$\Rightarrow v = \sqrt{\frac{8 \sin 30^\circ \times 0.7 \sin 30^\circ}{0.050}} = 5.3 \text{ m s}^{-1}$$



11. A planet of mass M and radius R rotates so rapidly that loose material at the equator just remains on the surface. What is the period of rotation of the planet?

G is the universal gravitational constant.

A $2\pi\sqrt{\dfrac{R}{GM}}$ **B** $2\pi\sqrt{\dfrac{R^2}{GM}}$ **C** $2\pi\sqrt{\dfrac{GM}{R^3}}$ **D** $2\pi\sqrt{\dfrac{R^3}{GM}}$

Answer: D

Gravitational force on material of mass m provides the centripetal force,

$$\frac{GMm}{R^2} = mR\omega^2$$

$$\frac{GM}{R^2} = R\left(\frac{2\pi}{T}\right)^2$$

$$T = 2\pi\sqrt{\frac{R^3}{GM}}$$

12. The following data refer to two planets.

	radius / km	density / kg m^{-3}
Planet P	2000	6000
Planet Q	16000	3000

The gravitational field strength at the surface of P is 13.4 N kg^{-1}.

What is the gravitational field strength at the surface of Q?

A 3.4 N kg^{-1} **B** 13.4 N kg^{-1} **C** 53.6 N kg^{-1} **D** 80.4 N kg^{-1}

Answer: C

gravitational field strength $g = \dfrac{GM}{R^2} = \dfrac{G}{R^2}\left(\dfrac{4}{3}\pi R^3\rho\right) = \dfrac{4\pi G\rho}{3}R \Rightarrow g \propto \rho R$

so, $\dfrac{g_Q}{g_P} = \dfrac{\rho_Q}{\rho_P} \times \dfrac{R_Q}{R_P} = \dfrac{3000}{6000} \times \dfrac{16000}{2000} = 4 \Rightarrow g_Q = 4 \times 13.4 = 53.6 \text{ N kg}^{-1}$

13. In a mixture of two monatomic ideal gases X and Y in thermal equilibrium, a molecule of Y has twice the mass of a molecule of X. The mean random translational kinetic energy of the molecules of Y is 6.0×10^{-21} J.

What is the mean random translational kinetic energy of the molecules of X?

A 3.0×10^{-21} J

B 4.2×10^{-21} J

C 6.0×10^{-21} J

D 12×10^{-21} J

Answer: C

- X and Y are in thermal equilibrium, so thermodynamic temperature is the same
- since mean k.e. per molecule is proportional to temperature, mean k.e. of molecules of X is the same as Y

14. The specific latent heat of vaporization of water at 20°C is appreciably greater than the value of 100°C.

This is because

A The molecules in the liquid are more tightly bound to one another at 20°C than at 100°C.

B More work must be done in expanding the water vapour against atmospheric pressure at 20°C than at 100°C.

C The root mean square speed of the vapour molecules is less at 20°C than at 100°C.

D The specific latent heat at 20°C includes the energy necessary to raise the temperature of one kilogram of water from 20°C to 100°C.

Answer: A

$$\Delta U = q + w \Rightarrow \Delta U = mL_v + w$$

Consider 1 kg of water, $L_v = \Delta U - w$

Quantity 1: <u>Change in internal energy ΔU</u>

- internal energy is sum of translational k.e. and p.e. of molecules.
- k.e. is proportional to thermodynamic temperature
- p.e. is the separation of molecules (molecules are closer at lower temperature)
- so, increase in internal energy to break the bonds for 20 °C water is higher (temperature remains same so no change in k.e.)

Quantity 2: <u>Work done w</u>
external pressure is the same so work done to expand is same

B: Incorrect. Refer to quantity 2
C: Incorrect. Temperature remains same during vapourisation so no impact
D: Incorrect. Vapourisation can happen at any temperature, no need to increase the temperature.

15. An ideal gas, contained in a cylinder by a frictionless piston, is allowed to expand from volume V_1, at a pressure p_1, to volume V_2, at pressure p_2. Its temperature is kept constant throughout. The work done by the gas is

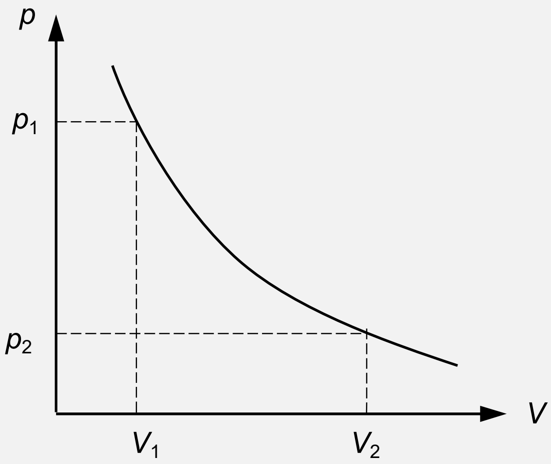
A Zero, because it obeys Boyle's Law and therefore $p_2V_2 - p_1V_1 = 0$.

B Negative, because the pressure has decreased and so the force on the piston has been diminishing.

C Zero, because it has been kept at constant temperature and so its internal energy is unchanged.

D Positive, because the volume has increased.

Answer: D



work done = area under p–V graph from V_1 to V_2

A: Incorrect. Although gas obey Boyle's law, area under graph ≠ 0
B: Incorrect. Gas expand indicate work done by gas is positive.
C: Incorrect. Although temperature is constant during expansion, area under graph ≠ 0

Note: When T is constant, $\Delta U = 0$ for ideal gas, so $q = -w \Rightarrow q =$ work done by gas

16. A body moves with simple harmonic motion of amplitude A and frequency $\frac{b}{2\pi}$.

What is the magnitude of the acceleration when the body is at maximum displacement?

A zero **B** $4\pi^2Ab^2$ **C** Ab^2 **D** $\dfrac{4\pi^2A}{b^2}$

Answer: C

$a = -\omega^2 x$ for SHM, so magnitude of acceleration $= \omega^2 x$

At maximum displacement A, acceleration $= (2\pi f)^2 A = \left(2\pi \times \dfrac{b}{2\pi}\right)^2 A = Ab^2$

17. Which one of the following statements concerning forced vibrations and resonance is correct?

A An oscillating body that is not resonating will return to its natural frequency when the forcing vibration is removed.

B At resonance, the displacement of the oscillating body is 180° out of phase with the forcing vibration.

C A pendulum with a dense bob is more heavily damped than one with a less dense bob of the same size.

D Resonance can only occur in mechanical systems.

Answer: A

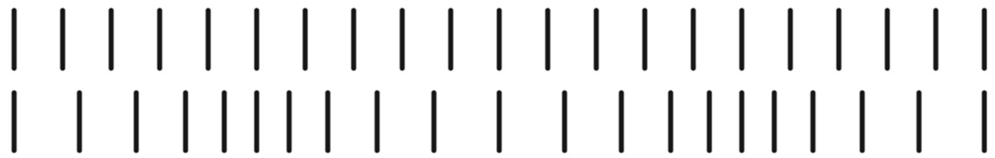
B: Incorrect. At steady state, displacement of forced oscillation is in–phase with the driving force.
If the driver is applied out–of–phase with the initial oscillation of the body, there will be a transient state of motion during which the body changes its motion to coincide with that of the driver.

C: This is a common misconception, mass ≠ damping.

D: Forced oscillations occurs in both mechanical and non–mechanical (e.g. electrical circuit such as inductor–capacitor circuit) systems.

18. The top row of bars represents a set of particles inside the Earth and at rest.

The lower row represents the same particles at one instant as a longitudinal wave passes from left to right through the Earth.



What should be measured to determine the amplitude of the oscillations of the particles in the lower row as the wave passes?

A half the maximum displacement of the particles from their position at rest

B half the maximum distance apart of the particles

C the maximum displacement of the particles from their position at rest

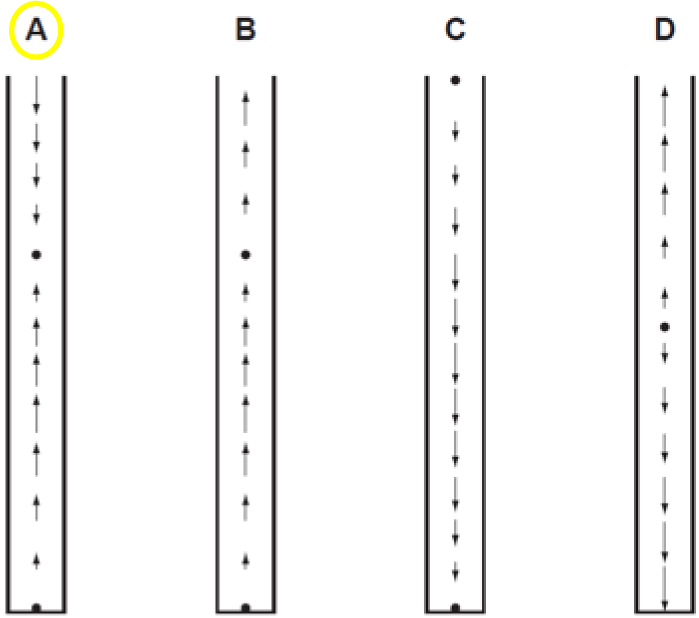
D the maximum distance apart of the particles

Answer: C

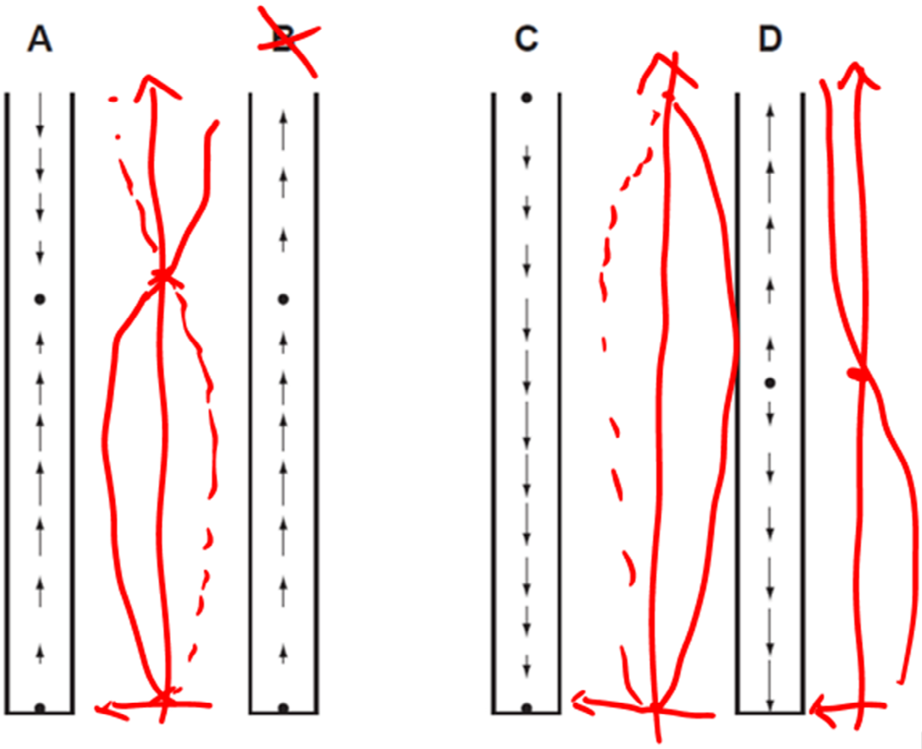
19. A stationary longitudinal wave is set up in a pipe.

In the diagram below, the length of each arrow represents the amplitude of the motion of the air molecules, and the arrow head shows the direction of motion at a particular instant.

Which diagram shows a stationary wave in which there are two nodes and two antinodes?



Answer: A



20. Light of wavelength 720 nm from a laser X is incident normally on a diffraction grating and a diffraction pattern is observed. Light from a laser Y is then also incident normally on the same grating. The third-order maximum due to laser Y is seen at the same place as the second-order maximum due to laser X.

What is the wavelength of the light from laser Y?

A 480 nm **B** 540 nm **C** 720 nm **D** 1080 nm

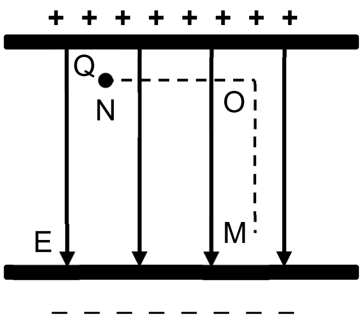
Answer: A

$d\sin\theta = n\lambda$

For same grating and at the same position, $d\sin\theta =$ constant

So, $2\lambda_X = 3\lambda_Y \Rightarrow \lambda_Y = \frac{2}{3}\times 720 = 480$ nm

21. The figure below shows a charged particle Q in a uniform electric field E.



Which of the following statements about Q is correct?

A No work is done when the positive charge moves from N to O then to M.

B No work is done when the positive charge moves from O to M only.

C Work is done by field when the positive charge moves from M to O, but work is done by external force when it moves from O to M.

D Work done on the positive charge when it moves from N to O to M is the same as when it moves from N to M directly.

Answer: D

Equipotential lines uniform electric field is perpendicular to the electric field lines, so
- no work done moving a charge from N to O or vice versa
- work is done moving a charge from O to M or vice versa

Potential at top > potential at bottom
- When positive charge moves from O to M, change in potential $\Delta V = V_M - V_O < 0$, so change in p.e. $= Q\Delta V < 0$. Decrease in p.e. indicate negative work done by external force or positive work done by the electric force (or field)
- When positive charge moves from M to O, work done by the external force

Work done on positive charge is independent of path taken but initial and final position.

22. Which one of the following statements about *electric field strength* and *electric potential* is **incorrect**?

A Electric potential is a scalar quantity.

B Electric field strength is a vector quantity.

C Electric potential is zero whenever the electric field strength is zero.

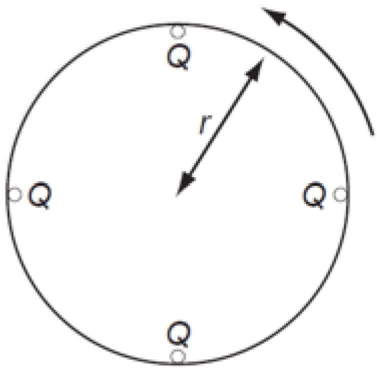
D The potential gradient is proportional to the electric field strength.

Answer: C

A: Correct. Electric potential is a scalar quantity.
B: Correct. Electric field strength is a vector quantity.
C: Incorrect. $E = -\frac{dV}{dx}$, so E is zero if and only if potential gradient is zero
D: Correct. $E = -\frac{dV}{dx}$

23. Four point charges, each of charge Q, are placed on the edge of an insulating disc of radius r.

The frequency of rotation of the disc is f.



What is the equivalent electric current at the edge of the disc?

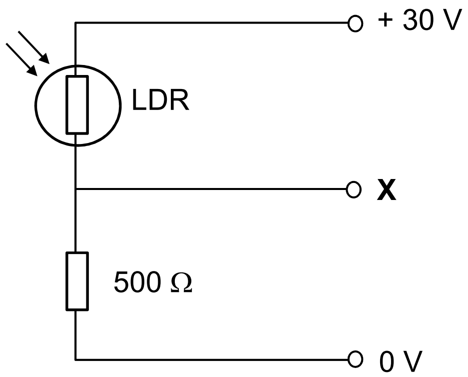
A $4Qf$ **B** $\frac{4Q}{f}$ **C** $8\pi rQf$ **D** $\frac{2Qf}{\pi r}$

Answer: A

Consider a fixed point, $4Q$ will pass through the point in one period.

So, $\text{current} = \frac{\text{total charge}}{\text{time}} = \frac{4Q}{1/f} = 4Qf$

24. The light dependent resistor (LDR) and a 500 Ω resistor form a potential divider between voltage lines held at +30 V and 0 V as shown.



The resistance of the LDR is 1000 Ω in the dark but drops to 100 Ω in bright light.

What is the corresponding change in the potential at **X**?

A A rise of 25 V

B A rise of 15 V

C A fall of 15 V

D A fall of 25 V

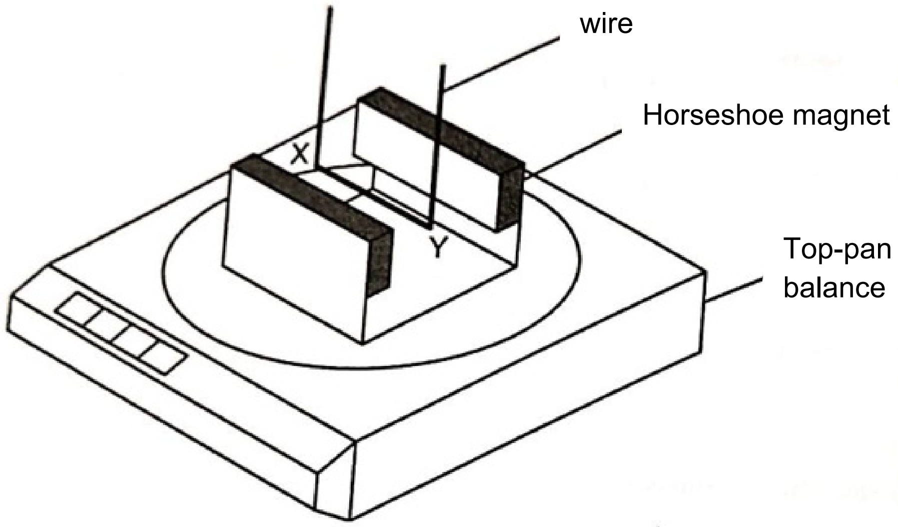
Answer: B

In dark, p.d. across LDR $= \dfrac{1000}{1000+500} \times 30 = 20$ V, so potential at X = +10 V

In bright light, p.d. across LDR $= \dfrac{100}{100+500} \times 30 = 5$ V, so potential at X = +25 V

Change in potential $= (+25) - (+10) = +15$ V

25. A horseshoe magnet rests on a top-pan balance with a wire situated between the poles of the magnet.



With no current in the wire, the reading on the balance is 142.0 g.

With a current of 2.0 A in the wire in the direction XY, the reading on the balance changes to 144.6 g.

What is the reading on the balance when there is a current of 3.0 A in the wire in the direction YX?

A 138.1 g **B** 140.7 g **C** 145.9 g **D** 148.5 g

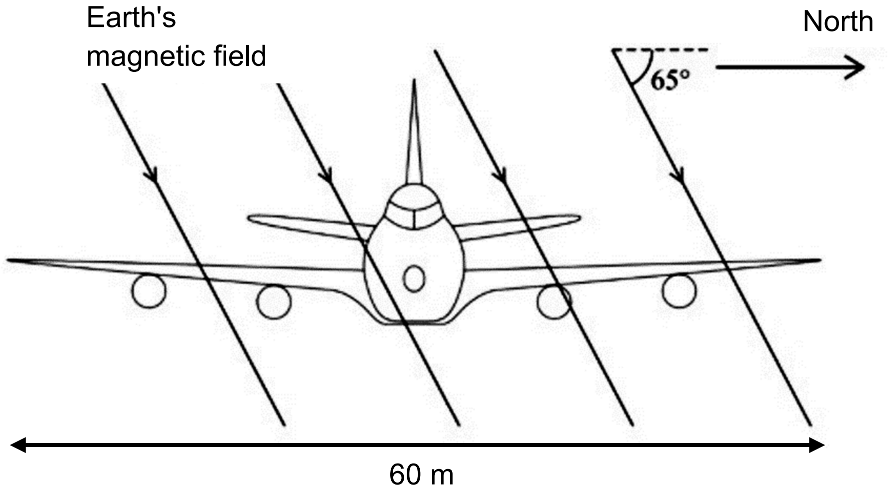
Answer: A

Force on XY is equal and opposite to the force exerted on magnet (Newton's 3rd law)
Force on XY, $F = BIL \Rightarrow F \propto I$

Without current, balance is measuring mass of magnet which is 142.0 g.
With 2.0 A in XY, force on magnet causes an increase in mass $= 144.6 - 142.0 = 2.6$ g

With 3.0 A in YX, force on magnet causes a decrease in mass $= 142.0 - \frac{2.6}{2} \times 3 = 138.1$ g

26. An aircraft, of wing span 60 m, flies horizontally at a speed of 150 m s^{-1}. The Earth's magnetic field in the region of the plane is 1.0×10^{-5} T and at an angle of 65° to the horizontal as shown below.



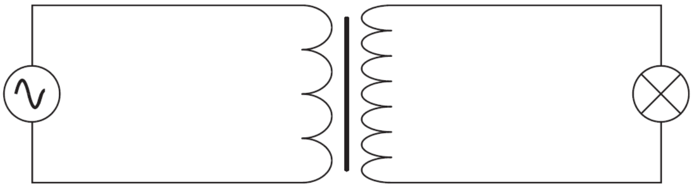
What emf is induced across the wing tips of the plane?

A 0.038 V **B** 0.082 V **C** 0.090 V **D** 9.0 V

Answer: B

Magnetic field perpendicular to wings $= 1.0 \times 10^{-5} \times \sin 65°$
Induced e.m.f. $= BLv = (1.0 \times 10^{-5} \sin 65°) \times 60 \times 150 = 0.082$ V

27. The primary coil of a step-up ideal transformer is connected to a source of alternating potential difference. The secondary coil is connected to a lamp.



Which line, **A** to **D**, in the table correctly describes the flux linkage and current through the secondary coil in relation to the primary coil?

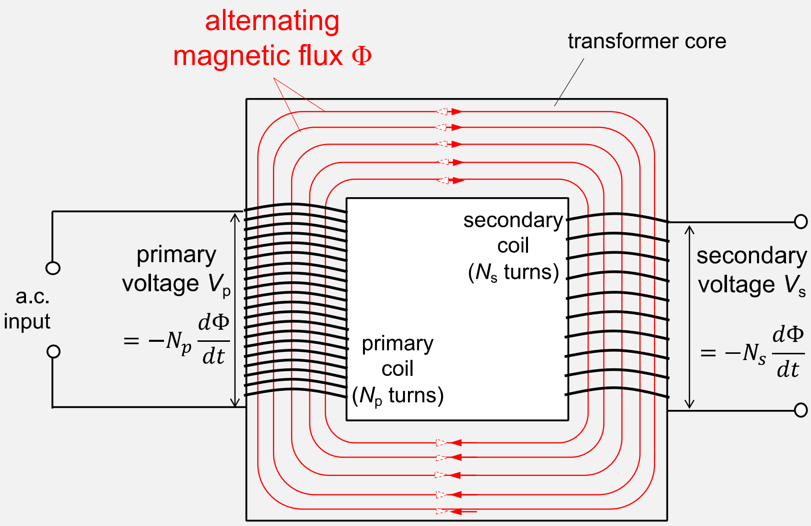
	$\dfrac{\text{secondary magnetic flux linkage}}{\text{primary magnetic flux linkage}}$	$\dfrac{\text{secondary current}}{\text{primary current}}$
A	> 1	< 1
B	< 1	< 1
C	> 1	> 1
D	< 1	> 1

Answer: A

$$\frac{N_S}{N_P} = \frac{V_S}{V_P} = \frac{I_P}{I_S}$$

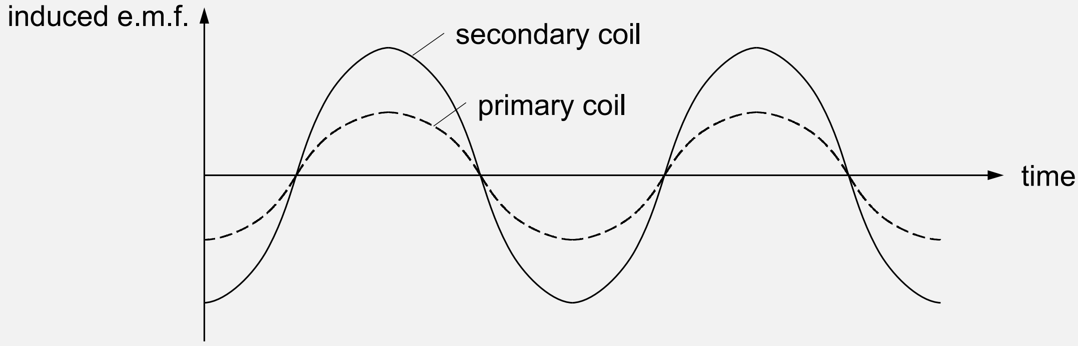
For a step-up transformer, $N_S > N_P$. So, $V_S > V_P$ and $I_P > I_S$

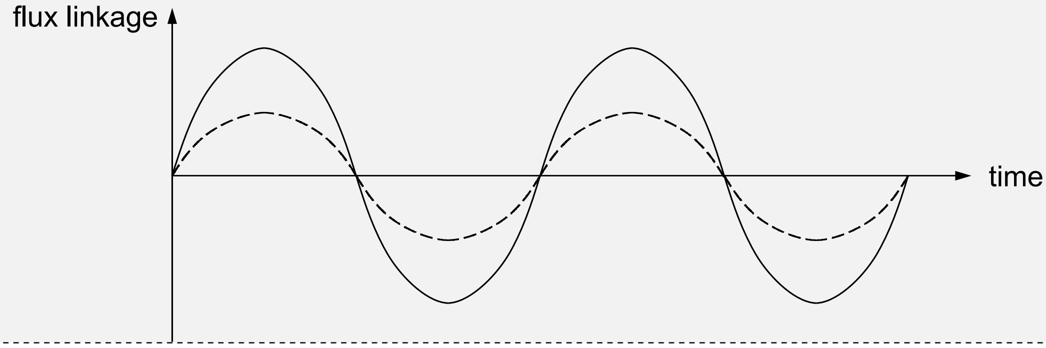
Consider the ideal transformer,



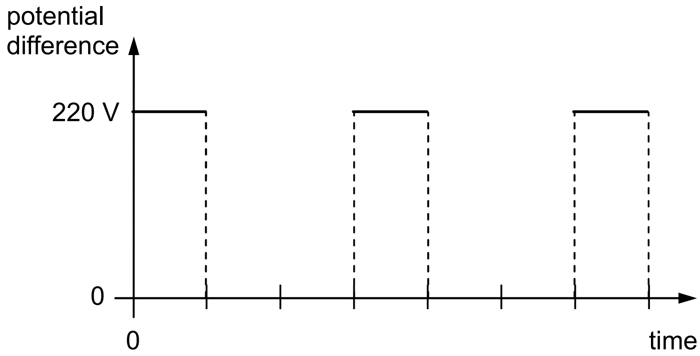
- alternating current in primary coil gives rise to an <u>alternating magnetic flux</u> Φ in the core <u>in phase</u> with the current in the primary coil
- since the flux is changing, by law of EMI, an e.m.f. will be induced in the primary coil is $V_P = -N_P \dfrac{d\Phi}{dt}$ and the secondary coil is $V_S = -N_S \dfrac{d\Phi}{dt}$
- so, $V_P = -\dfrac{d}{dt}(N_P\Phi)$ and $V_S = -\dfrac{d}{dt}(N_S\Phi)$ where $N\Phi$ is flux linkage

- Since $V_S > V_P$, flux linkage of secondary coil > primary coil (see below)





28. An alternating current is supplied to a rectifier. The output from the rectifier has the shape shown in the diagram, with a maximum value of 220 V.



This output is used as the supply to a resistor of resistance 60.0 Ω.

What are the r.m.s. values of the current and the power to the resistor?

	r.m.s. current / A	power / W
A	1.22	89.6
B	1.83	202
C	2.12	269
D	2.59	403

Answer: C

r.m.s. voltage $= \sqrt{\dfrac{220^2 \times \frac{1}{3}T}{T}} = 127.02\text{ V}$

r.m.s. current $= \dfrac{127.02}{60} = 2.117\text{ A}$

power $= I_{\text{r.m.s.}}{}^2 R = 2.117^2 \times 60 = 269\text{ W}$

29. When a beam of light of wavelength λ is incident normally on a plane mirror, it is reflected totally.

The intensity of the light corresponds to a rate of n photons per second.

What is the force exerted on the mirror by the beam of light?

A $nh\lambda$ **B** $\dfrac{nh}{\lambda}$ **C** $2nh\lambda$ **D** $\dfrac{2nh}{\lambda}$

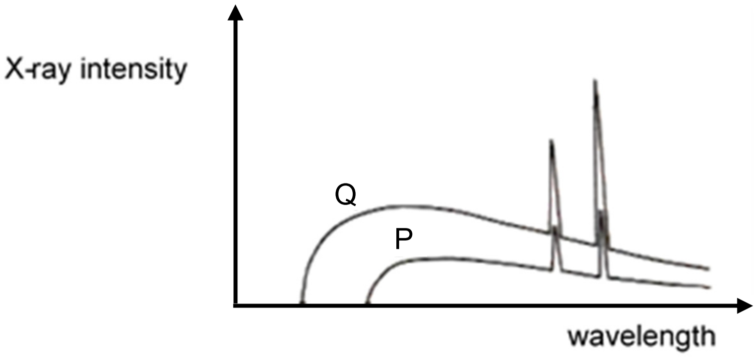
Answer: D

momentum of a photon $p = \dfrac{h}{\lambda}$

change in momentum when photon is reflected completely $= p - (-p) = \dfrac{2h}{\lambda}$

force exerted $=$ rate of change of momentum of photons $= \dfrac{\text{no. of photons}}{\text{time}} \times \dfrac{2h}{\lambda} = \dfrac{2nh}{\lambda}$

30. The given graph show the variation of X-ray intensity with wavelength emitted from two X-ray tubes P and Q.



Which of the following could be a deduction from the graphs?

A X-ray tube Q has the higher voltage applied to it and the target material in both tubes is the same.

B X-ray tube Q has the higher voltage applied to it and the target material in both tubes are different.

C X-ray tube P has the higher voltage applied to it and the target material in both tubes is the same.

D X-ray tube P has the higher voltage applied to it and the target material in both tubes are different.

Answer: A

The characteristic x-ray is the same (spikes are at the same wavelengths) so target material must be the same.

Q has a smaller minimum wavelength than P. Minimum wavelength when all k.e. of incident electron is converted to a single photon, so

$$e \times \text{accelerating p.d.} = \frac{hc}{\lambda_{\text{min}}}$$

$$\lambda_{\text{min}} = \frac{hc}{e \times \text{accelerating p.d.}}$$

Therefore, accelerating p.d. of Q is higher than P.

2020 Prelim H2 Physics Paper 2 (Suggested Solutions)

1	**(a)**		for a body in (rotational) equilibrium, sum of/total clockwise moments about a point = sum of/total anticlockwise moments about the same point	B1
	(b)	**(i)**	**1.** (Taking moment about P,) $3.4 \times 10^4 \times 5 = T \times 12$ T = 14 200 N *correct substitution for all terms*	M1 A0
			2. moment of vertical rope = $T \cos 24^{\circ} \times 2.5$ (Taking moment about P,) $T \cos 24^{\circ} \times 2.5 = F \times 8$ T = 4050 N *correct substitution for all terms*	C1 M1 A0
		(ii)	Assume that mass of beam is negligible / light $R_Y = 14200 + 4050\cos 24.0^{\circ} = 17900$ N $R_X = 4050\sin 24.0^{\circ} = 1647$ N $\alpha = \tan^{-1}\dfrac{R_X}{R_Y}$ $\alpha = 5.26^{\circ}$, $R = 18000$ N	C1 C1 A1
	(b)		trailer accelerates forward, require horizontal component of force vertical component of force remains the same, so magnitude of F increases resultant of the components at an angle to the vertical towards the car	B1 B1 B1
			[Total: 10]	

2	**(a)**			The hot gases exit (the rear of) the rocket with momentum The rocket must (momentarily) acquire an equal and opposite momentum resulting in the forward thrust.	B1 B1
	(b)			There is no drag force due to air resistance when the rocket is in vacuum When in air, drag force due to air resistance opposes the motion of the rocket, leading to a smaller resultant force.	B1 B1
	(c)	**(i)**		thrust (upwards, longer arrow) weight (downwards, shorter arrow)	B1 B1
		(ii)		$\text{thrust} - mg = ma$ $\text{thrust}\left(= ma + mg\right) = 1.2 \times 10^6 \times \left(3.5 + 9.81\right)$ $= 1.60 \times 10^7 \text{ N}$	M1 B1
		(iii)		$\text{thrust} - mg = ma \Rightarrow \quad m(g+a) = \text{thrust} = 1.60 \times 10^7$ $m = \dfrac{1.60 \times 10^7}{9.81 + 18.7} = 561207 \text{ kg}$ $\dfrac{\Delta m}{\Delta t} = \dfrac{1.2 \times 10^6 - 561207}{60} = 1.06 \times 10^4 \text{ kg s}^{-1}$	C1 A1
				[Total: 10]	

3	**(a)**	**(i)**		For the car to **<u>stay in contact with the floor</u>, the normal contact force N > 0.** $mg - N = \frac{mv^2}{r}$ $N = mg - \frac{mv^2}{r}$ $N > 0$ $mg > \frac{mv^2}{r}$ $v > \sqrt{gr} > \sqrt{9.81 \times 12.0}$ $v > 10.8 \text{ m s}^{-1}$	B1 M1 A1
		(ii)		By conservation of energy, lost in gravitational p.e. = gain in kinetic energy (energy conservation) $mg(2 \times 12) = \frac{1}{2}mv^2 - \frac{1}{2}m(10.850)^2$ $v = 24.3 \text{ m s}^{-1}$	M1 A1
	(b)	**(i)**		angle turned per unit time *or* rate of change of angle *or* rate of change of angular displacement in a particular direction	M1 A1
		(ii)		if the car is below minimum speed in **(a)(i)** (or at rest), its **<u>weight</u>** will cause it to fall inwards arm can exert an **<u>upward</u>** force (equal to weight) on the carriage in **(b)** even when it is at rest	B1 B1
					[Total:9]

4	**(a)**		work done per unit positive charge in bringing a small test charge from infinity to that point	B1
	(b)	**(i)**	p.d. = 15 V	A1
		(ii)	change in potential $\Delta V = 25 - 5 = 20$ V work done $= q\Delta V = 5 \times 10^{-6} \times 20$ $= 1.0 \times 10^{-4}$ J	 M1 A1
		(iii)	The potential increase positively as the charge move from B to D. Since it is a positive charge, work has to be done by external agent to move it to a more positive potential line.	B1 B1
	(c)	**(i)**	electric field strength at a point is numerically equal to the potential gradient at that point	B1
		(ii)	towards the +20 V equipotential line and approximately perpendicular at the point.	B1
		(iii)	move away to the right / towards lower potential / away from higher potential initial acceleration is small, acceleration is (much) larger beyond the +20 V line	B1 B1
			[Total: 10]	

5	**(a)**			magnetic force per unit current per unit length acting on a current-carrying long straight conductor placed at right angle to the magnetic field.	B1
	(b)			**charged particle (electrons)** moving **perpendicularly to magnetic field** experience force indicate direction towards face QRGF e.g. P to Q, E to F	B1 B1
	(c)	**(i)**		deflected electrons **build–up** on one face so **lower potential** than opposite face	M1 A1
		(ii)		Face QRGF and face PSHE	A1
	(d)	**(i)**		$\text{density} = \dfrac{2.7\times10^{-3}}{\left(1\times10^{-2}\right)^3} = 2.7\times10^{3}\text{ kg m}^{-3}$ $\text{number per unit volume} = \dfrac{2.7\times10^{3}}{4.5\times10^{-26}}$ $= 6.0\times10^{28}\text{ m}^{-3}$	C1 M1 A0
		(ii)		$V_H = \dfrac{0.15\times4.6}{6.0\times10^{28}\times0.090\times10^{-3}\times1.60\times10^{-19}}$ $= 8.0\times10^{-7}\text{ V}$	M1 A1
				[Total: 10]	

[Turn over

6	**(a)**	**(i)**		$\text{current} = \dfrac{\text{power}}{\text{p.d}} = \dfrac{6.0}{24} = 0.25\text{ A}$	A1
		(ii)		$\text{p.d.} = \dfrac{24}{12} = 2.0\text{ V}$	A1
		(iii)		$\text{resistance} = \dfrac{\text{p.d}}{\text{current}} = \dfrac{2.0}{0.25} = 8.0\ \Omega$	B1
	(b)	**(i)**		Break occurs in the connection between P, switch, battery and Q.	B1
		(ii)		Break occurs in light bulb H / filament of H has melted/broken	B1
		(iii)		Break occurs somewhere along the wires connecting the bulbs	B1
	(c)	**(i)**		$\text{p.d.} = \dfrac{24}{11} = 2.2\text{ V}$	A1
		(ii)		$\text{new power} = \dfrac{V^2}{R} = \dfrac{2.182}{8} = 0.5951\text{ W}$ original power = 0.50 W $\text{fractional increase} = \dfrac{0.5951 - 0.50}{0.50} = 0.19$	C1 A1
				[Total: 9]	

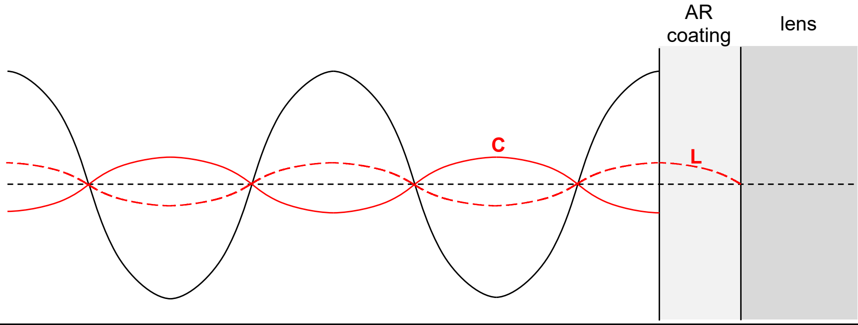
7					
	(a)	**(i)**	[Method 1] power $=$ intensity $\times$ area so $R = \dfrac{\text{intensity of reflected wave} \times \text{area}}{\text{intensity of incident wave} \times \text{area}} = \dfrac{\text{intensity of reflected wave}}{\text{intensity of incident wave}}$ intensity $\propto$ amplitude^2 so $R = \dfrac{k \times (\text{reflected wave amplitude})^2}{k \times (\text{incident wave amplitude})^2} = \left(\dfrac{\text{reflected wave amplitude}}{\text{incident wave amplitude}}\right)^2$ since r is the ratio of amplitudes, so $R = r^2 = \left(\dfrac{n_2 - n_1}{n_2 + n_1}\right)^2$ [Method 2] intensity $\propto$ amplitude$^2 \Rightarrow$ intensity $= k \times$ amplitude^2 power $=$ intensity $\times$ area $= k \times$ area $\times$ amplitude^2 $R = \dfrac{k \times \text{area} \times (\text{reflected wave amplitude})^2}{k \times \text{area} \times (\text{incident wave amplitude})^2} = \left(\dfrac{\text{reflected wave amplitude}}{\text{incident wave amplitude}}\right)^2$ since r is the ratio of amplitudes, so $R = r^2 = \left(\dfrac{n_2 - n_1}{n_2 + n_1}\right)^2$		B1 B1 (B1) (B1)
		(ii)	**1.**	sum of reflected and transmitted power (or energy) equal incident power (or energy)	B1
			2.	$T + R = 1$ or $T = 1 - R$ $T = \dfrac{\left(n_2^2 + n_1^2 + 2n_2n_1\right) - \left(n_2^2 + n_1^2 - 2n_2n_1\right)}{\left(n_2 + n_1\right)^2}$ $T = \dfrac{4n_2n_1}{\left(n_2 + n_1\right)^2}$	M0 A1
		(iii)	**1.**	at air–lens interface, $R = \left(\dfrac{1.5 - 1.0}{1.5 + 1.0}\right)^2 = 0.04$ and $T = 0.96$ at lens–air interface, $T' = 0.96 \times 0.96 = 0.9216$ *percentage of transmitted light* $= 0.9216 \times 100\% \approx 92\%$	C1 M1
			2.	*at air–lens interface,* $R = \left(\dfrac{1.58 - 1.0}{1.58 + 1.0}\right)^2 = 0.050538$ *and* $T = 0.94946$ *at lens–air interface,* $T' = 0.94946 \times 0.94946 = 0.90147$ *percentage decrease* $= (1 - 0.90147) \times 100\% = 9.85\%$	 C1 A1
	(b)	**(i)**	(from Fig. 7.4, reflection is minimised when $n_c < n_{\text{lens}}$,) reflected waves from air–coating and coating–lens interfaces undergo same 180° phase change		B1

(wave in coating travels additional half a wavelength,) both reflected waves have half a wavelength path difference — M1

both reflected waves superpose destructively to give minimum/zero displacement/ amplitude — A1

Note: Technically the thickness of the coating is related to the refractive index. However, the question is simplified because optical path difference is not stated explicitly in H2 Physics learning outcome.

(ii) correct shape for C (180° phase change) with amplitude < incident — M1
correct shape for L (180° phase difference from C) with amplitude < incident — A1
amplitude of **L** equal/slightly less than **C** (award if first two marks are awarded) — A1



(c) **(i)**

refractive index n of lens	optimum refractive index n_c of the coating
1.50	1.2250
1.55	$1.2440 \le n_c \le 1.2460$
1.60	$1.2640 \le n_c \le 1.2650$
1.65	$1.2840 \le n_c \le 1.2850$
1.70	$1.3035 \le n_c \le 1.3045$

All answers expressed to 4 d.p. (1/2 smallest square) — A1

(ii) $n_c = k\sqrt{n} \Rightarrow k = n_c / \sqrt{n}$

n	n_c	k
1.50	1.2250	1.00021
1.55	1.2450	1.00001
1.60	1.2645	0.99968
1.65	1.2845	0.99998
1.70	1.3040	1.00012

use at least three sets of data to verify — M1

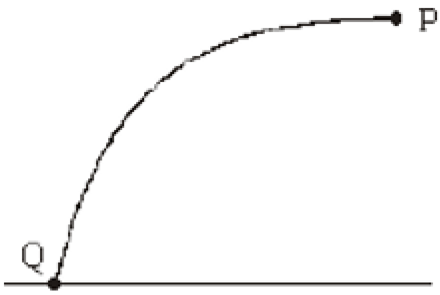
since k agrees to 3 s.f. (or 4 s.f. depending on data), n_c is proportional to square root of n — A1

Only 1 mark awarded if two sets of data used.

(iii) from (ii), $k \sim 1.000$
so, $n_c = \sqrt{1.58} = 1.26\,(1.257,\ 1.2570)$ — A1

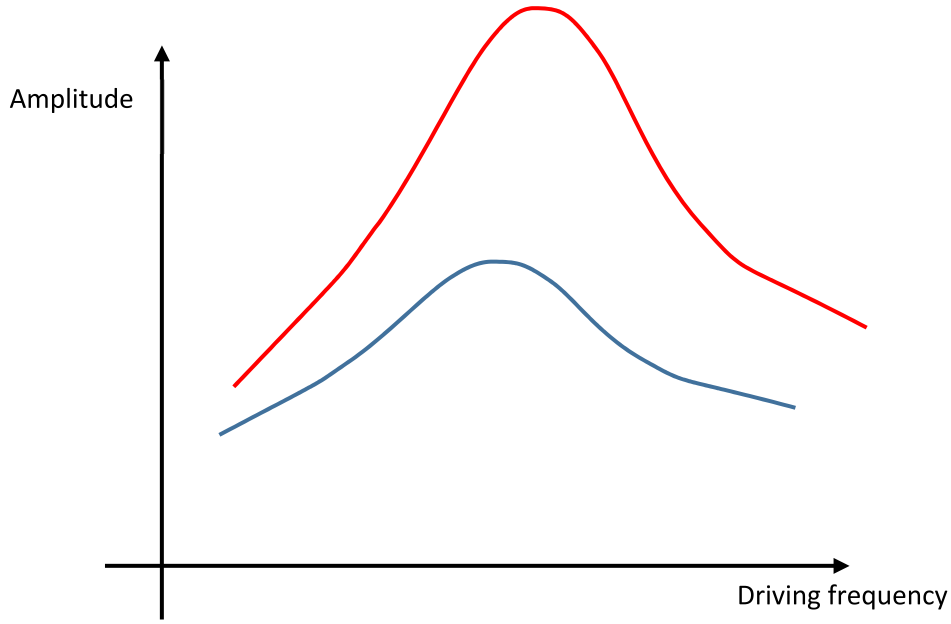
	(d)	(i)	visible light comprises of a range of / different wavelengths reduce reflection of these wavelengths through different thickness / refractive index of coating (since T and R depends on refractive index)	M1 A1
		(ii)	obtain thickness of one layer by using one wavelength from the of visible light spectrum between 400 nm and 700 nm obtain thickness of 7 layers (14 layers if coating both sides considered) with value within the range $0.7\ \mu\text{m} \le \text{thickness} \le 2.4\ \mu\text{m}$ (minimum $= 7 \times \frac{1}{4} \times 400 \times 10^{-9} = 0.7\ \mu$m, maximum $= 2 \times 7 \times \frac{1}{4} 700 \times 10^{-9} = 2.4\ \mu\text{m}$)	M1 A1
				[Total: 22]

2020 SH2 H2 Physics Prelim P3 Suggested Solution

1	(a)				B1
	(b)			No **horizontal force** acting on it. Hence, no horizontal acceleration.	M1 A0
	(c)	(i)		$v^2 = u^2 + 2as$ $32^2 = 0 + 2\ (9.81)s$ s = 52.2 m = 52 m	C1 A0
		(ii)		Time taken to travel 52 m, $s = ut + \frac{1}{2} a t^2$ (OR $v = u + at$, $v = 32$) $52 = 0 + \frac{1}{2}(9.81)t^2$ t = 3.26 s QR = 3.26 (95) = 310 m	C1 A1
	(d)			<u>Horizontal speed is reduced more significantly due to the much higher velocity.</u> <u>Vertical resultant force</u> is <u>reduced</u>. Vertical acceleration is less. Time is longer ⇨ right of Q	M1 A1

[Turn over

2	(a)	Gravitational potential energy is taken to be zero at infinity and gravitational forces are attractive. Since the external agent is applying a force in the opposite direction relative to the displacement made by the mass, negative work is done by the external agent.		[B1] [B1]
	(b)	(i)	At infinity, kinetic energy and gravitational potential energy of probe is zero. By conservation of energy, the minimum kinetic energy required to for probe from surface to reach infinity is GMm/R_E Since E is smaller than minimum required Kinetic energy, the probe cannot travel to deep space. Thought Process: - Calculate/determine the minimum energy required for probe to travel to infinity from surface. - Compare the minimum energy required with the energy of probe. Comment Note: At surface, KE + GPE = 0 (Assume very far) By COE, Energy on Earth = Energy at infinity $KE_s + GPE_s = 0$ $KE_s = -GPE = GMm/R_E$	[M1] [A1]
		(ii)	By conservation of energy Energy at surface = Energy at that point Initial KE + Initial GPE = final GPE + final KE $\frac{3}{4} GMm/R_E + (-GMm/R_E) = (-GMm/R) + 0$ $-\frac{1}{4R_E} = -\frac{1}{R}$ R = 25.7 x 10^6 m OR Loss in KE = Gain in GPE $\frac{3GMm}{4R_E} = \frac{GMm}{R_E} - \frac{GMm}{R}$, note $R_E < R$ R = 25.7 x 10^6 m	[C1] [B1] [A1]
		(iii)	Earth rotate in the West to East direction. Space probe launch in this direction will have an initial velocity, thus it has some initial KE, and not zero.	[B1] [A1]

3		(i)	Resonance	[B1]
		(ii)	the bridge can swing on its own natural frequency. When the frequency of the forced oscillation due to walking pedestrians matches the natural frequency of the bridge, (resonance occurs). The amplitude may increase and damage the bridge structure.	[B1] [B1] [B1]
		(iii)	 **Fig. 1.1**	
			Correct shape and amplitude. Since it is before damping, the amplitude must be higher. Damping will make the period longer, thus frequency is lower. So before damping, the frequency must be higher.	 [B1] [B1]

4	(a)			transfers energy in direction of wave propagation by oscillations (of particles in the medium) profile of the wave appears to be moving although the particles in the medium do not get transported along the wave.	M1 A1
	(b)	(i)		two wavelengths = 0.70 m wavelength = 0.70 / 2 = 0.35 m	A1
		(ii)		4.5 cycle takes 5 ms period = 5 / 4.5 ms frequency = 1 / period = 900 Hz	A1
		(iii)		$\text{speed} = 900 \times 0.35$ $= 315 \text{ m s}^{-1}$	A1
	(c)			$\text{intensity} \propto \dfrac{1}{\text{distance}^2} \Rightarrow \text{intensity} = \dfrac{k_1}{\text{distance}^2}$ $\text{intensity} \propto \text{amplitude}^2 \Rightarrow \text{intensity} = k_2 \times \text{amplitude}^2$ $\text{so amplitude} = \sqrt{\dfrac{k_1}{k_2}} \times \dfrac{1}{\text{distance}} = \dfrac{k}{\text{distance}}$	C1

possible data and value for *k*

x / m	amplitude / mm	*k* / 10^{-4} m^2
0.17 or 0.18	1.60	2.72 or 2.88
0.34 or 0.35	0.80	2.72 or 2.8
0.52 or 0.53	$0.50 < \text{amplitude} < 0.55$	~2.7
0.70	0.40	2.8
~0.87	~0.30	~2.6
1.05	~0.25	~2.6

M1

k is approximately equal, so relationship is obeyed — A1

Allow conclusion to be based on the calculations with 2 sets of data although it is recommended that more than 2 sets of data should be used.

5	(a)			$\rho = R\frac{A}{L} = R\frac{\pi(\frac{d}{2})^2}{L}$ $= 4.8(\frac{\pi(\frac{0.23x10^{-3}}{2})^2}{0.40} = 4.9857x10^{-7}\ \Omega\,m$ $\frac{\Delta\rho}{\rho} = \frac{\Delta R}{R} + \frac{\Delta L}{L} + 2\frac{\Delta d}{d}$ $= \frac{0.05}{4.80} + \frac{0.1}{40.0} + 2\frac{0.01}{0.23} = 0.99873$ $\Delta\rho = 0.99873\text{x } 4.957x10^{-7} = 4.979 \text{ x } 10^{-8}$ $= 5 \text{ x } 10^{-8}\ \Omega\,m$ OR Maximum $\rho = 4.85(\frac{\pi(\frac{0.24x10^{-3}}{2})^2}{0.399} = 5.4990x10^{-7}\ \Omega\,m$ Minimum $\rho = 4.75(\frac{\pi(\frac{0.22x10^{-3}}{2})^2}{0.401} = 4.5028\ x10^{-6}\ \Omega\,m$ $\Delta\rho$ = ½(5.4990- 4.5028) x10^{-7} = 4.980 x 10^{-8} = 5 x 10^{-8} Ω m	C1 C1 A1 C1 C1 A1
	(b)	(i)		Energy is dissipated in the internal resistance. Total energy is supplied by source is distributed to the external circuit and the internal resistance. Hence, electrical energy per unit charge available for external circuit is reduced.	M1 A1
		(ii)		Calculation of R_x, $\frac{1}{4.8} = \frac{1}{12} + \frac{1}{R_x}$ $R_x = 8.0\ \Omega$ Let V be the pd across the thermistor. Ratio= $\frac{\frac{V^2}{R_x}}{\frac{V^2}{12}} = \frac{\frac{V^2}{8}}{\frac{V^2}{12}} = \frac{12}{8} = 1.5$	C1 B1 A1
		(iii)		As temperature decreases resistance of thermistor X increases Total circuit resistance increases, total current decreases as EMF is constant. Power decreases.	B1 M1 A0

[Turn over

	(b)	**(i)**		Power of lamp exceed the power rating of the filament which is 60 W. The <u>filament had broken</u> and the <u>circuit has become open.</u> Note: When V = 15.4 V, power of lamp = 15.4(5.9)= 90.86 W 90.86/60 = 1.5 times	**B1**
		(ii)		Resistance of the lamp is defined as the ratio of the pd across the lamp and the current passing through the lamp.	**B1**

6	(a)	The magnetic flux ϕ is defined as the product of the area and the magnetic flux density that is perpendicular to it.		[B1]
	(b)	(i)	$\Phi_N = NBA \sin\theta$ $= 650\,(2.5 \times 10^{-3})\,(20)(35)(10^{-3})^2 \sin 30^o$ $= 5.69 \times 10^{-4}$ wb turns	[M1] [M1] [A1]
		(ii)	$E = N\Delta\Phi/t$ $= N\,(\Phi_f - \Phi_i)\,/\,t$ $= 0 - 5.69 \times 10^{-4} / (5 \times 10^{-3})$ $= 0.114$ V	[M1] [A1]
		(iv)	$0.114 = I(5)$ $I = 0.0228$ A $Q = It = 0.0228 \times 5 \times 10^{-3}$ $= 114 \times 10^{-6}$ C	[C1] [A1]

7	(a)			increasing current causes increasing flux (in core) (according to Faraday's law of induction,) induces <u>e.m.f. in coil</u> (from Lenz's law, induced e.m.f.) opposes the increase of current (in circuit)	**B1** **B1** **B1**
	(b)			constant maximum current gives constant flux / field induced e.m.f. only when flux (in core/coil) is changing (so, no induced e.m.f.)	**B1** **B1**
	(c)	(i)		$\frac{N_S}{N_P} = \frac{V_S}{V_P}$ r.m.s. voltage of output $= \frac{52}{\sqrt{2}} (= 36.77\text{ V})$ $N_S = \frac{52/\sqrt{2}}{90} \times 1200 = 490$ (*accept 491 turns*)	**C1** **A1**
		(ii)		0 ms or 7.5 ms or 15.0 ms or 22.5 ms	**A1**

8	**(a)**			CLT not tested	

9	(a)	(i)	Explain what is meant by the *internal energy of an ideal gas*.
			It is the sum of random distribution of kinetic energies of the gas molecules B[1]
		(ii)	Mean random translational kinetic energy of ideal gas molecules = $\frac{3}{2}kT$ For a system containing N molecules of ideal gas, the internal energy = total random kinetic energy $= N \times \frac{3}{2}kT = \frac{3}{2}NkT$ M[1] Since pV = NkT Internal energy $= \frac{3}{2}pV$ A[1]
	(b)	(i)	The molar mass of carbon dioxide is 44.0 g. Show that the mass of one molecule of carbon dioxide is about 7.3 x 10^{-26} kg. Mass of one molecule $= \frac{44.0 \times 10^{-3}}{6.02 \times 10^{23}}$ M[1] $= 7.31 \times 10^{-26}$ kg
		(ii)	The surface temperature of the planet Mercury is 467 °C. Determine the root-mean-square speed of one molecule of carbon dioxide at this temperature. Average k.e. per molecule: $\frac{1}{2}mc_{rms}^2 = \frac{3}{2}kT$ $\frac{1}{2}(7.31 \times 10^{-26})c_{rms}^2 = \frac{3}{2}(1.38 \times 10^{-23})(467 + 273)$ C[1] $c_{rms}^2 = 419{,}097$ C[1] $c_{rms} = 647$ ms^{-1} A[1]
		(iii)	The root mean square speed of one carbon dioxide molecule is less than the escape speed. M[1] Hence, an atmosphere of carbon dioxide can exist on Mercury's surface. A[1]

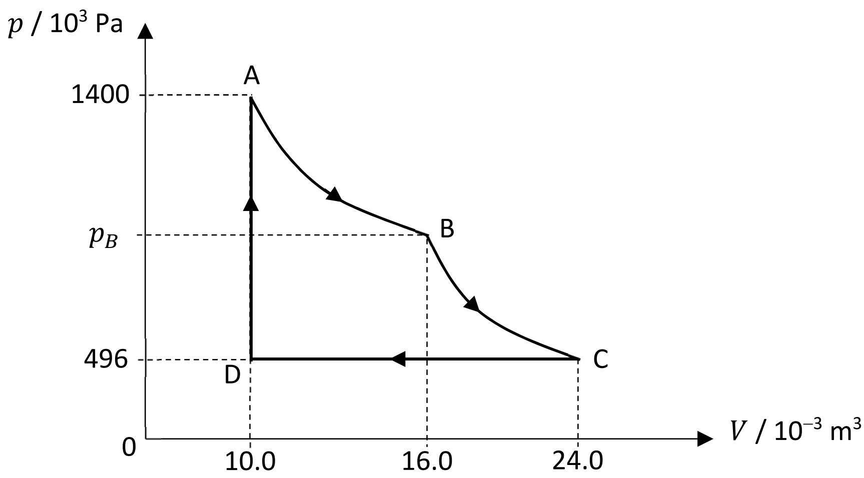
9 (c) (i) State the first law of thermodynamics.

The increase in internal energy B[1]

of a system is equal to the

sum of heat supplied and work done on the system. B[1]

(ii) In the pV diagram shown below (not to scale), an ideal gas expands isothermally from point A to point B and then adiabatically from point B to point C where its volume is 24.0 x 10^{-3} m^3. An isobaric compression brings it to point D where its volume is 10.0 x 10^{-3} m^3. An isochoric process returns the gas to point A.



(1) Show that the pressure of the gas at point B is 875 x 10^3 Pa.

$p_B V_B = p_A V_A$ $\quad p_B = \frac{p_A V_A}{V_B} = \frac{(1400\times10^3)(0.010)}{0.016}$ M[1]

$= 875$ kPa

(2) Complete the following table.

	q / joules	w / joules	ΔU / joules
A → B	6580 [1]	– 6580	0
B → C	0 [1]	– 3140	– 3140
C → D	–17340	6940 [1]	– 10 400

[3]

(3) $\Delta U_{D\ to\ A} - 3140 - 10\ 400 = 0$ C[1]

$\Delta U_{D\ to\ A} = 13\ 540$ J A[1]

				or $\frac{3}{2}(1400-496)\times10^{3}\times10\times10^{-3}$ =13 560 J
		(iii)	• Gas molecules <u>transfer kinetic energy</u> to the <u>receding piston</u> during collision M[1] • Resulting in a decrease in the average kinetic energy of the gas molecules A[1] • Since average (random) kinetic energy of the gas molecules is proportional to (thermodynamic) temperature of the gas M[1] • Gas temperature decreases A[0]	

10	(a)	According to wave theory, emission of electrons from the surface of the metal can occur at any frequency B[1] If exposure time is sufficiently long since energy can be accumulated B[1] For particulate model, electromagnetic radiation comes in discrete quantum/packet, where energy of each quantum/packet/photon proportional to frequency of radiation. B[1] Emission of electrons occur only if photon energy is greater than work function B[1]		
	(b)	(i)		$E_k = e\Delta V = 1.60 \times 10^{-19} \times 4700 = 7.52 \times 10^{-16}$ J C1 $E_k = \frac{1}{2}mv^2 = \frac{p^2}{2m}$ C1 $\lambda = \frac{h}{\sqrt{2mE_k}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.11 \times 10^{-31} \times E_k}}$ M1 $\lambda = 1.8 \times 10^{-11}$ m A1
		(ii)		De Broglie wavelength of electrons is of the same order of magnitude as the spacing between atoms in a crystal B1 Electron diffraction pattern reveals the arrangement of atoms within the crystal B1
	(c)	(i)		A photon is emitted when an electron transits from a higher energy level to a lower one. B1 Discrete lines showed that only photons of specific energies are emitted B1 Specific energy change implies discrete energy levels B1
		(ii)	(1)	Arrow from -1.51 eV to -3.41 eV B1
			(2)	$\lambda = \frac{hc}{(3.41 - 0.38)e}$ C1 $\lambda = 4.10 \times 10^{-7}$ m A1
		(iii)		continuous spectrum crossed by dark lines B1 four dark lines B1 corresponding to wavelengths of photons absorbed by hydrogen atom B1 absorbed photons re-emitted in all directions B1